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T. Douglas Price *Editor*

Isotopic Proveniencing and Mobility

The Current State of Research

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Editor

Isotopic Proveniencing and Mobility

The Current State of Research

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ISSN 1568-2722 ISSN 2730-6984 (electronic)
Interdisciplinary Contributions to Archaeology
ISBN 978-3-031-25721-6 ISBN 978-3-031-25722-3 (eBook)
https://doi.org/10.1007/978-3-031-25722-3

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Cover illustration: Kelly Knudson preparing samples for analysis in the Laboratory for Archaeological Chemistry at the University of Wisconsin-Madison (Photo: T. Douglas Price).

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The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Preface

Isotopic proveniencing involves the isotopic characterization (O, Sr, and Pb) of tooth enamel or petrous bone. Applications in archaeology involve both humans and animals. Isotopic ratios of these elements vary geographically. As enamel and petrous bone form in early childhood and often do not change, they retain the chemical signature of the place of birth. Comparison with isotopic ratios from the place of burial allows one to distinguish local from non-local individuals. Sulfur isotopes in bone convey information on coastal versus inland provenience and can be useful in studies of adult mobility. The combination of isotopic proveniencing and aDNA has moved archaeological interest in migration and mobility to the fore, but there is very little overview published for either methodology for archaeologists and bioarchaeologists.

I have several things to say about isotopic proveniencing, and I realized that this could best be done in an edited volume with other papers on the subject. This volume should provide state-of-the-art presentation and discussion of procedures, especially what works and what doesn't—what's been learned over the last 30 years that isotopic proveniencing has been underway in archaeology. There is no current consensus on best practice. The contributors have been selected from among the leaders in the field, those with active research and hands-on experience with the approach. Moreover, focus in the volume will be on application, not method, to emphasize to the reader the wide range of questions that can be addressed using isotopic proveniencing. A series of suggestions and recommendations conclude the volume.

Isotopic proveniencing has become an almost standard procedure in the analysis of archaeological burials as a means of distinguishing locals from foreigners. The number of publications has undergone at least a tenfold increase in the last 20 years. At the same time, the field has evolved and new procedures and guidelines have emerged that have not been widely disseminated. New directions emerge that are not always fruitful. This volume aims at state-of-the-art examples and discussion of the new directions and limitations that apply to isotopic proveniencing, including topics such as samples, baselines, isoscapes, distinguishing non-locals, and place of origin.

The hype cycle is perhaps a useful way to characterize the field of isotopic proveniencing (Fenn & Raskino, 2008; Jones & Bösl, 2021). The hype cycle describes the lifespan of an idea or innovation as a series of highs and lows associated with scientific or technological opportunities. This cycle moves from an initial trigger to a peak of expectations, often followed by a trough of disillusionment, then a rising slope of enlightenment and a plateau of productivity. This cycle in archaeology has certainly characterized radiocarbon dating and aDNA research. It would seem that isotopic proveniencing fits this description as well and may be entering a phase of disillusionment at the moment. My own perspective, hopefully demonstrated in this volume, is certainly positive. The field has enormous potential and some problems to be solved.

I would especially like to thank the individuals who have contributed to this volume for their time and input. Bioarchaeology is a rapidly growing field, and the demands on one's time and energy are unlimited relative to what most of us can do. It is the thought and effort of the contributors that gives this volume any consequence that it may have.

I must also thank the staff of Springer, and especially Jelmer Eerkens, the editor of the *Interdisciplinary Contributions to Archaeology* series, and Christi Lue, associate editor for Archaeology & Logic & Cognition. I would also thank the reviewers of the papers and the book. There is no question that your input improved this volume.

Madison, WI, USA T. Douglas Price
November 2022

References

Fenn, J., & Raskino, M. (2008). *Mastering the hype cycle: How to choose the right innovation at the right time.* Harvard Business Press.

Jones, E. D., & Bösl, E. (2021). Ancient human DNA: A history of hype (then and now). *Journal of Social Archaeology, 21*, 236–255.

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Chapter 9
Application of Gaussian Mixture Model Clustering for the Resolution of Multi-isotope Fingerprints



Andrea Göhring, Gisela Grupe, and Peer Kröger

Abstract The analysis of multi-isotope fingerprints has become common during the last decade, and there is no doubt that this will be the future of this field in archaeometry. However, isotopic fingerprints are very complex not only because of their heterogeneity. They are also prone to some uncertainty, mainly because of their contextual proxy character. Also dimensionality matters according to the number of isotope systems chosen. When coping with multidimensional isotope data, a promising interpretation of multi-isotope fingerprints is cluster analysis. In this chapter, we present three examples of successful applications of Gaussian Mixture Model (GMM) clustering to 4- to 6-dimensional data sets established for the scope of provenance analysis in very complex eco-geographical settings.

Benefits of GMM clustering compared to other common methods such as PCA or k-means are the allocation of an individual sample to its cluster with a mathematical probability and the potential of identifying individual samples with mixed isotopic signals due to, for example, intra-life migration. The latter is a specific problem of bone analysis that can often not be avoided (e.g., in cremations). As is the case for all modeling and computer-aided procedures, it cannot be emphasized enough that while also GMM quickly and always results in a clustering outcome, the results necessitate a careful and in-depth interpretation with regard to the research question(s) and the archaeological context.

Keywords Diet · Gaussian Mixture Model clustering (GMM) · Knowledge Discovery in Databases (KDD) · Mobility · Multi-isotope fingerprint · Palaeoecology · Provenance

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T. D. Price (ed.), *Isotopic Proveniencing and Mobility*, Interdisciplinary Contributions to Archaeology, https://doi.org/10.1007/978-3-031-25722-3_9

9.1 Introduction

To decipher the past by analyzing stable isotopic signatures in the bodily remains of the former witnesses (archaeological skeletons) as detailed as possible, the analysis of multi-isotope fingerprints has become common during the last decade, and the relevant publications leave no doubt that this will be the future of this field in archaeometry. The term "multi-isotope fingerprint" describes a pattern detected in multi-isotope data which is characteristic for a specific region or an individual. This multi-isotope pattern can be used to "fingerprint," for example, the origin of geological or biological materials as well as the habitat, water sources, or dietary preferences of a specimen (Bowen et al., 2009; Grupe et al., 2017; Mauder et al., 2017). Nearly every stable isotope ratio tells a different story. While $\delta^{15}N$ and $\delta^{13}C$ in bone collagen are related to dietary protein, $\delta^{13}C$ in the structural carbonate is derived from all major dietary components including the energy part, namely, carbohydrates and fat. $\delta^{34}S$ helps in distinguishing marine from terrestrial diets and is particularly meaningful in coastal areas, but in such ecosystems, the influence of the sea spray effect needs to be taken into account (Göhring et al., 2018). $\delta^{18}O_{phosphate}$ depends on hydrological cycles and relates to climate, precipitation, and drinking habits, while $\delta^{18}O_{carbonate}$ is more sensitive to the oxygen isotopic composition of consumed food (Kirsanow & Tuross, 2011). Bioavailable strontium with its $^{87}Sr/^{86}Sr$ ratio is primarily related to geological parameters, soil composition, and precipitation (Lengfelder et al., 2019). Lead stable isotopes ($^{208}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, $^{206}Pb/^{204}Pb$) are also geodependent, typical for mining areas (ore deposits), but can influence local isotopic signatures of more distant regions by pollution (e.g., Haack et al., 2003; Artioli et al., 2016; Shaw et al., 2016; Price et al., 2017). However, even in the most recent literature, multi-isotope fingerprints are still conservatively interpreted one by one or by bivariate plots (e.g., Jay et al., 2013; Madgwick et al., 2019; Lafoon et al., 2019; Scorrer et al., 2021). This way, palaeodiet, provenance, and habitat preference are dealt with separately.

The first questions to address are therefore: What information shall be sought and which isotopic signature is appropriate for supporting or rejecting a scientific hypothesis? Analysis of one or two isotopic ratios (e.g., $\delta^{13}C$ combined with $\delta^{15}N$ for the reconstruction of palaeodiet or $^{87}Sr/^{86}Sr$ combined with $\delta^{18}O$ for questions related to mobility) and univariate statistical analyses usually lead to gross assessments that, however, frequently remain on the descriptive level. As soon as stable lead isotopes are considered, the lead isotopic signature of a skeleton has already three dimensions.

The question arises how to better cope with and decipher a multi-isotope fingerprint. Isotopic fingerprints are very complex not only because of their heterogeneity. They are also prone to some uncertainty, first because of the measurement errors but mainly because of their contextual proxy character (e.g., dependency from metabolic parameters). Also dimensionality matters according to the number of isotope systems chosen. A promising interpretation of multi-isotope fingerprints is cluster analysis. In general, "clustering is the task of assigning a set of objects into groups

(called clusters) so that the objects in the same cluster are more similar (in some sense or another) to each other than to those in other clusters" (Drivelos & Georgiou, 2012). Under the assumption that individuals that shared the same environment had similar feeding habits and similar places of origin, these individuals should exhibit similar isotopic fingerprints. Cluster analysis therefore permits a similarity search among groups of individuals. Apparent dissimilarities need to be evaluated with regard to the probable underlying reasons. In this chapter, we demonstrate the research potential of cluster analysis in the frame of Knowledge Discovery in Databases (KDD). While we will restrict our experience to stable isotopes and provenance analysis, it shall be at least mentioned that first applications to multiple osteological measurements have also led to encouraging results (Göhring, 2020b; Göhring et al., 2020a).

9.1.1 Knowledge Discovery in Databases and Gaussian Mixture Model Clustering

KDD concerns the whole data analysis process, starting with the raw data set and a research question and ending up with a gain of knowledge. The process is interactive as well as iterative, based on several decisions made by the user. This way, knowledge is the end product of a nontrivial data-driven analysis. A thorough application of all different steps of the KDD process, including preprocessing of the data, data analysis, and interpretation of the results, is essential to gain useful knowledge. Otherwise, patterns without any meaning may result (Fayyad et al., 1996a, b; Piatetsky-Shapiro, 1990).

The target data set is generated based on the research question(s). Where necessary, the data are preprocessed or transformed. Adjusted to the goal, specific data mining methods (e.g., cluster analysis) are selected. An explorative analysis follows and both the model and the hypothesis are chosen. Moreover, data mining algorithms and suitable models and parameters are determined. The aim of the data mining step is the detection of patterns in the data set that give clues to the solution of the research question(s). It might be necessary to repeat one or more of the preceding steps (Fayyad et al., 1996a; Fayyad, 1996; Frawley et al., 1992).

Cluster analysis is a useful data mining method for pattern recognition. It aims to detect clearly separable subgroups in the data set whereby neither the number of subgroups nor group-specific features need be known beforehand (Fraley & Raftery, 1998, 2002). The identified subgroups should be cohesive and separated from other groups (Fraley & Raftery, 2002). Thus, the data points within each group should be as similar to each other as possible, while different groups should be as different as possible (Aggarwal, 2014, 2015; Xiong & Li, 2014).

A novel method derived from the KDD procedure is the Gaussian Mixture Model (GMM) clustering that has been applied to multi-isotope fingerprints several times and proved superior to common methods such as Principal Component Analysis

(PCA; Mauder et al., 2019). GMM cluster analysis is based on multivariate normal distributions. Each cluster is represented by a parametric probability distribution (here: Gaussian distribution). This allows grouping those data points into one cluster which – with a certain probability – belong to the same distribution (Aggarwal, 2015; Deng & Han, 2014). GMM cluster analysis is an EM (Expectation Maximization)-based multivariate clustering method, whereby the EM algorithm determines the maximum likelihood for, for example, the GMM model. The algorithm uses two iterative steps. During the E-step, the posterior probability of a cluster allocation is determined based on the given parameter set (Aggarwal, 2014; Deng & Han, 2014). In the course of the subsequent M-step, the mean vector is calculated. Based on these new values, both covariances and mixing probabilities can be recalculated (Deng & Han, 2014). The logarithmized likelihood is calculated and subsequently maximized. Both E- and M-steps are iteratively performed until the algorithm converges against an optimal parameter set (Aggarwal, 2015; Deng & Han, 2014).

Each GMM cluster is ellipsoid and centered around a mean vector. Geometric features, such as volume, shape, and orientation, are determined by the covariance matrix (Fraley & Raftery, 1998; Scrucca et al., 2016). A variety of different model types is possible in the R package "mclust" (Scrucca et al., 2016). In the case of the simplest model, the covariance matrix is a multiple of the identity matrix I. All components are spherical and of equal size (mclust model "EEE"; Scrucca et al., 2016). The most complex model allows varying volume, size, and shape (mclust model "VVV"; Scrucca et al., 2016).

The optimal cluster model is chosen on the basis of the Bayesian Information Criterion (BIC), which is based on an asymptotic approximation of the transformation of the Bayesian posterior probability of the chosen model (Reddy & Vinzamuri, 2014). While the BIC is usually defined in such a way that it is minimal for an optimal model (see, e.g., Reddy & Vinzamuri, 2014), it is defined differently in the "mclust" R package using an opposite sign in order to make an interpretation of BIC plots easier (Fraley & Raftery, 1998). Therefore, in the case of "mclust", BIC is maximal for an optimal clustering model (Fraley & Raftery, 1998; Scrucca et al., 2016).

The underlying uncertainty of group membership for each data point can be additionally calculated. The latter is assessed through conditional probabilities obtained from the EM algorithm (Bensmail et al., 1997; Fraley & Raftery, 1998). Data points with a high attributed uncertainty are more likely misclassified (Fraley & Raftery, 2002) as they could be allocated to several clusters with a certain probability.

The applicability of GMM cluster analysis for the interpretation of multidimensional isotope fingerprints has repeatedly been demonstrated (Göhring, 2019; Göhring et al., 2016, 2018, 2020a, c; Grupe et al., 2018, 2020; King et al., 2021; Mauder, 2017; Mauder et al., 2016a, b; Tschetsch et al., 2020).

Feature ranking can be used to quantify the information content of different isotopic systems. Accordingly, feature ranking allows detecting an isotopic system with high and another with low information content. While a multidimensional

isotopic fingerprint is usually preferable, limited sample material, i.e., in archaeological contexts, could necessitate omitting one or more isotopic systems in order to reduce the required sample material for analyses. Feature ranking thus permits the identification of those isotopic systems which can be omitted without losing too much information (Göhring et al., 2020b). In other cases, the information hidden in a certain isotopic system may even cause a worse clustering outcome than for the data set excluding the respective isotopic system (Mauder et al., 2019) when it is unable to differentiate between groups, while the remaining systems in the data set can detect those groups. In some cases, the inefficient system does not have any impact on the clustering results. In such a case, a data set omitting the disturbing variable is advisable.

In this publication, we perform feature ranking using the Adjusted Rand Index (ARI) that is based on a reference cluster analysis of a control group. ARI compares two partitions of data by a pairwise comparison of their clustering conformity (Hubert & Arabie, 1985). Two special cases of ARI, the structural relevance and the structural redundancy, are especially meaningful. Relevance describes to what extent a single feature (here: isotopic system) can reproduce the clustering structure of the control group. Redundancy, on the other hand, explains to what extent the control group structure can be reproduced after removal of one feature (Mauder et al., 2017). The maximum value of ARI is 1, indicating an identical clustering result. An ARI of 0 implies a random behavior. If the correspondence between two cluster results is lower than expected, ARI is below 0 (Mauder et al., 2017).

9.2 Material and Methods

9.2.1 Material

Three example data sets are presented to demonstrate the explanatory power of KDD and GMM cluster analysis for deciphering multi-isotope fingerprints in archaeological bones and teeth.

Example 9.1 Transalpine mobility – coping with a spatial redundancy of single isotope ratios.

GMM clustering aiming at deciphering a multi-isotope fingerprint was first attempted in the frame of the interdisciplinary project "Transalpine Mobility and Culture Transfer", financially supported by the German Science Foundation (www.for1670-transalpine.uni-muenchen.de; Grupe et al., 2017). Contextual focus was on provenance analysis of archaeological cremated human and animal remains recovered in the region of the European Alps. Since the Brenner Pass is the lowest one in the Alps with an altitude of only 1370 m asl, a north-to-south transect via this pass had been in use since the Mesolithic (Metzner-Nebelsick et al., 2017). $^{87}Sr/^{86}Sr$ ratios alone do not work as transalpine mobility cannot be assessed for archaeological finds

because the northern and southern alpine forelands are both dominated by carbonate soil. The inneralpine areas are dominated by older crystalline rock, however with a very small-scale variability. Also, marine limestone is frequently evidenced due to the genesis of the mountains in the place of the former Tethys Sea. Therefore, a multi-isotope fingerprint was chosen for an isotopic mapping of this transect including $\delta^{18}O$ besides $^{87}Sr/^{86}Sr$ with regard to the vertical stratification of the region and lead stable isotopes because of the rich ore deposits that had been sought after since the Bronze Age. Initially, a multi-isotope fingerprint of bioapatite consisting of these isotopic ratios was established for 217 individual compact bone samples from local cattle, domestic pigs, and red deer. The bones had been sampled from 30 sites along this transalpine route, whereby the site distribution in the three gross regions (north, inneralpine, south) was uneven for archaeological reasons. Samples per site varied between 3 (one site only) and 12. The raw data including the conventional isotopic map by plotting the isotope ratios one by one was published by Toncala et al. (2017; see also quality control and apatite purification protocols). GMM experiments were performed to test whether clusters recognize the known location of the excavation sites.

Example 9.2 Birds from the Jutland peninsula – defining migratory species, habitat preferences, and hunting grounds.

Wild and domesticated birds from the Viking Age Haithabu and medieval Schleswig sites in Northern Germany had been analyzed in terms of $\delta^{13}C_{collagen}$, $\delta^{15}N_{collagen}$, $\delta^{13}C_{carbonate}$, and $\delta^{18}O_{carbonate}$ (original data published by Doppler et al., 2010). The sites constitute a settlement continuum on the Jutland peninsula. The bird data set includes 177 individuals of two domesticated and 23 wild species. Bone collagen stable isotope ratios were accepted for interpretation when the C/N_{molar} ratio fulfilled the criteria published by DeNiro (1985) and Harbeck and Grupe (2009). $\delta^{18}O_{carbonate}$ needs to be viewed with caution because of its susceptibility to diagenetic alteration. However, the finds were exceptionally well preserved, and stable oxygen isotope ratios in carbonate followed every expectation (Doppler et al., 2010). The region under study is ecologically very complex because the peninsula separates the Baltic Sea from the Atlantic and terrestrial, freshwater, marine, and brackish water environments exist in closest vicinity. Mixed isotopic ratios occur on a regular base, and uni- or bivariate analyses will soon reach their limits. GMM experiments were performed to test whether clusters recognize habitat and feeding preferences as well as migratory behavior.

Example 9.3 Provenance analysis and detection of human intrapopulation differentiation by combining collagen and apatite stable isotope analyses.

$\delta^{13}C_{collagen}$, $\delta^{15}N_{collagen}$, $\delta^{13}C_{carbonate}$, $\delta^{18}O_{carbonate}$, $\delta^{34}S_{collagen}$, and $^{87}Sr/^{86}Sr$ had been measured in the bones and teeth of 36 adult humans from the same sites as in Example 9.2 (parts of the original data are published by Göhring et al. (2020c), Grupe et al. (2013); complete data set is available in Göhring (2019)). Both sites were busy trade centers at their time; therefore, population admixture is expected. The location of the

site in a region that is overprinted by the last glaciations is geologically difficult because of the co-existence of local carbonate soil and boulder clay introduced by the advancing ice masses. This example illustrates the information gained by combining GMM clustering with isotopic sourcing, in this case by the use of the SISUS (Stable Isotope Sourcing Using Sampling) mixing model (Erhardt, 2014).

9.2.2 Methods

9.2.2.1 Gaussian Mixture Model (GMM) Cluster Analysis

GMM cluster analysis was performed by the "GMMbuilder" tool (Mauder et al., 2016c) (Example 9.1) and the R package "mclust" (version 5.4.5; Scrucca et al., 2016; Examples 9.2 and 9.3). A user-friendly R script for the application of GMM using "mclust" is available online (Göhring, 2020a). The only prerequisite for the application is the installation of the software tool R (R Core Team, 2021) and (optional) RStudio, the integrated development environment for R (RStudio Team, 2020). By small changes to the R script (e.g., number of isotopic systems, cluster number, cluster model), the script can be easily adjusted to another data set.

9.2.2.2 Feature Ranking

Feature ranking based on ARI was determined using the "adjustedRandIndex" command of the "mclust" package (Scrucca et al., 2016). For structural relevance, the classification of a single variable is compared with the control group. In the case of the redundancy of a variable, the respective variable is removed from the data, and cluster results of this subset are compared with the control group (see Mauder et al., 2017).

9.2.2.3 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) with scaled variables was conducted using the R package "stats" (R Core Team, 2021). PCA biplots with group labels can be created using the R packages "ggplot2" (Wickham, 2016) and "ggfortify" (Horikoshi & Tang, 2016).

9.2.2.4 Stable Isotope Mixing Models Using SISUS

Several different approaches for the calculation of stable isotope mixing models have been developed, including both frequentist (probabilistic) and Bayesian approaches (for a review of several models see, e.g., Grupe, 2014; Phillips et al.,

2014). The algorithm of the probabilistic program SISUS (stable isotope sourcing using sampling) first determines the vertices and borders of the solution polygon of the mixing model (Erhardt, 2014) using the so-called double description method (Fukuda, 2005; Fukuda & Prodon, 1996). In the subsequent step, a probabilistic sample is chosen from the solution polygon (Erhardt, 2014). For more details on the algorithm, see Erhardt (2014) and Smith (1984). The R package "sisus" (version 3.9-14) allows the calculation of the stable isotope mixing model using the SISUS approach (Erhardt, 2014).

9.3 Results

9.3.1 Example 9.1: Transalpine Mobility

Two GMM cluster experiments were performed with stable oxygen, strontium, and lead isotopes from 217 individual animal bones, both without any labeling of the individual bone samples according to the location of their excavation site. The first experiment resulted in four clusters labeled 0, 1, 2, and 3 (Table 9.1, Fig. 9.1a). Average probability of individual cluster assignment was $p = 0.94$. Cluster 0 was the largest one that captured 94 individuals. Animals assigned to this cluster are found almost everywhere along the transect, and even distant places of recovery are not differentiated (blue color in Fig. 9.1a). Cluster 1 was the smallest one and mostly found further to the north of the mountain chain but occasionally also in the inneralpine areas. The same holds for cluster 2, which is also rarely represented in the south. Cluster 3 was restricted to the south and the inneralpine areas. After this first experiment, location of the excavation sites was poorly reflected by the clusters. The only information that could be deduced from this first GMM experiment is that in case individuals ascribed to cluster 3 had been excavated to the north of the Alps, and individuals assigned to clusters 1 and 2 had been excavated to the south, then these individuals should have crossed the Alps, and migration direction is evident.

According to ARI, none of the five isotope ratios was of outstanding explanatory power what is easily explained by the spatial redundancies mentioned above. Not unexpected, $\delta^{18}O_{phosphate}$ exhibited the lowest structural importance in the data set (Mauder et al., 2019). In a follow-up study, Tschetsch et al. (2020) were able to show that for this particular data set, a better spatial allocation of the finds by the multi-isotope fingerprint became possible when the cluster assignment of an individual sample had a probability of at least 0.71. By omitting all individual samples with a lower cluster assignment, the data set was reduced by only 8% to $n = 198$. But still, $\delta^{18}O_{phosphate}$ tended to blur the multi-isotopic similarity rather than being of considerable explanatory power.

Therefore, the isotope fingerprint was reduced to four dimensions in the second GMM experiment, using the radiogenic isotope ratios only. With regard to the necessary assignment probability, the original data set was reduced by 11% to $n = 186$.

Table 9.1 Variability of apatite isotope ratios in the clusters generated by the first (O, Sr, Pb) and second (Sr, Pb) GMM experiment

O, Sr, Pb	**$\delta^{18}O_{phosphate}$**	**$^{87}Sr/^{86}Sr$**	**$^{208}Pb/^{204}Pb$**	**$^{207}Pb/^{204}Pb$**	**$^{206}Pb/^{204}Pb$**
c0 (n = 94)	−3.9	0.70708–0.71318	38.3381–38.7605	15.6262–15.6985	18.5811–19.4533
c1 (n = 25)	−3.2	0.70863–0.71658	38.1878–38.6898	15.6071–15.8570	18.6339–21.3966
c2 (n = 38)	−3.1	0.70820–0.71325	38.2927–38.5364	15.6043–15.6463	18.4210–18.9500
c3 (n = 41)	−4.6	0.71262–0.72068	38.2317–38.8265	15.5178–15.7048	18.3299–19.2494
Sr, Pb	**$\delta^{18}O_{phosphate}$**	**$^{87}Sr/^{86}Sr$**	**$^{208}Pb/^{204}Pb$**	**$^{207}Pb/^{204}Pb$**	**$^{206}Pb/^{204}Pb$**
c0 (n = 70)		0.70708–0.71654	38.2659–38.7769	15.6377–15.7113	18.5811–19.7480
c1 (n = 15)		0.70851–0.71591	38.3550–38.7221	15.6249–15.6962	18.5918–19.0877
c2 (n = 35)		0.70820–0.71701	38.2317–38.5364	15.5178–15.6710	18.3525–18.9500
c3 (n = 19)		0.70885–0.71658	38.1878–38.6898	15.6071–15.8570	18.6339–21.3966
c4 (n = 47)		0.70919–0.72068	38.4534–38.8265	15.6135–15.7048	18.4661–18.9383

This second experiment resulted in five clusters labeled 0–4, whereby the generation of an additional cluster already indicates a more detailed similarity search (Table 9.1, Fig. 9.1b). This four-dimensional GMM experiment led to much clearer results. Average individual cluster assignment was still very good with p = 0.90. Again, cluster 0 was the largest one with 70 individuals, but this time, it was geographically restricted to the north, which is characterized by carbonate soil and some inneralpine areas. Cluster 1 is exclusively found to the south of the transect where crystalline bedrock dominates. Cluster 2 is mainly found to the north of the Alps and also at two sites in the inneralpine areas. Cluster 3 is restricted to the north and only occasionally occurs in the inneralpine region, while cluster 4 is restricted to the inneralpine regions and the south. This time, GMM clusters reflect the known location of the excavation site and prove their research potential for provenance studies. It needs to be emphasized that any attempt for provenance analysis in this geologically complex region by the use of $^{87}Sr/^{86}Sr$ ratios alone needs to fail. The considerable variability of this isotope ratio along the alpine transect (Toncala et al., 2017: Fig. 1) is mirrored in the $^{87}Sr/^{86}Sr$ variability per cluster (Table 9.1). In both GMM experiments, the range extended from <0.709 until >0.713; therefore, lead stable isotopes were indispensable.

The second GMM experiments led to a much better spatial differentiation of the samples since, for example, both clusters 1 and 4 are not evidenced in the north at all. It is even logical that while cluster 0 dominates the north, also individuals assigned to clusters 2 and 3 are frequently found there. Geologically, this northern region is a patchwork of quaternary glacial river pavements and moraines (white

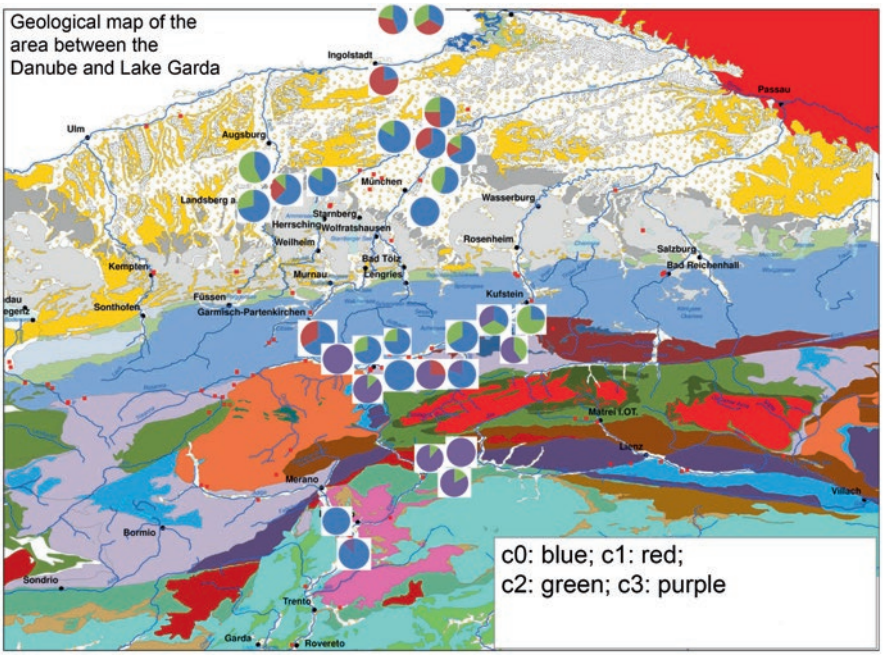


Fig. 9.1a Result of the GMM clustering with stable oxygen, strontium, and lead isotopic ratios. Pie charts are plotted on the excavation sites and indicate the proportion of animal bones captured by one of the four clusters. Original geological map by F. Söllner (Toncala et al., 2017). Bluish colors, calcareous soils and limestone; pink, red, and brown colors: crystalline and metamorphic sediments; and white and yellow colors, quaternary cover of the Alpine foreland with tertiary molasse, mainly sand and sandstone

and gray colors in Figs. 9.1a and 9.1b) and tertiary molasse (yellow). Not only the spatial allocation of the clusters but also the frequency of individuals captured by them permits an assessment of the place of origin combined with probabilities that is impossible by the use of one or two stable isotope ratios only. The cluster distribution indicates a clear north-to-south gradient across the Alps, and only a handful of samples are most probably imports to the site of their recovery (e.g., two animals from the inneralpine region that were captured by cluster 3, Fig. 9.1b).

What about the samples that were not assigned to any cluster with a sufficient probability, what applied to about 10% in both GMM experiments? One explanation could be that these individuals originated from a different and distant region and had been imported to their site of recovery. Since the bone had been analyzed, mixed isotope ratios might also be responsible. Knowledge of the approximate age at death of the individuals in question is requisite to test whether they had lived long enough to accumulate mixed isotopic signatures. This will frequently be the case in humans and constitutes a problem that is faced when cremations are analyzed. A GMM cluster analysis of strontium and lead stable isotopes in human cremations from the Fritzens-Sanzeno culture (5th/4th century BCE) identified both such mixed

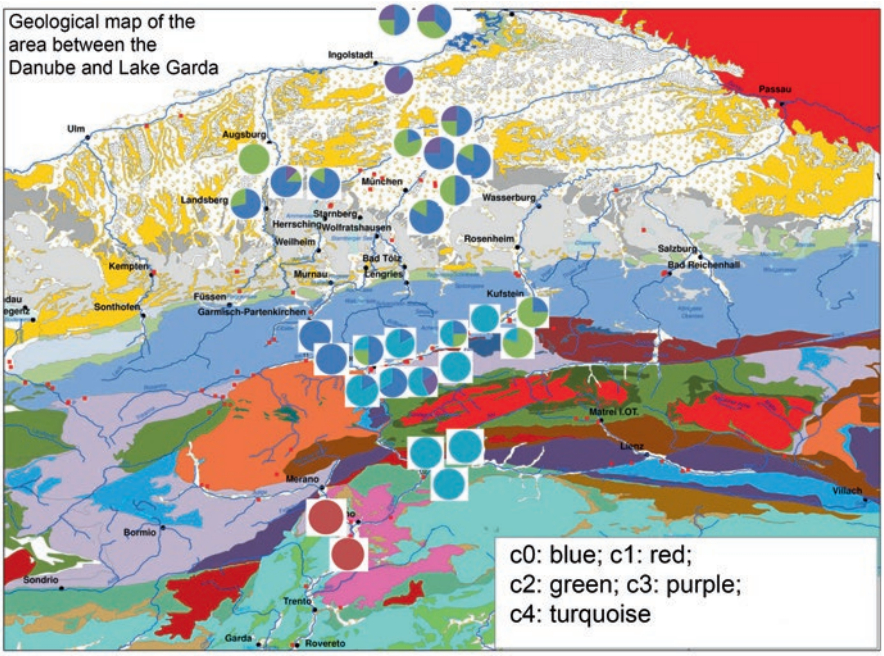


Fig. 9.1b Result of the GMM clustering with stable strontium and lead isotopes only. Pie charts are plotted on the excavation sites and indicate the number of animal bones captured by one of the four clusters. (Original geological map by F. Söllner (Toncala et al., 2017). Color code of the map (see Fig. 9.1a))

isotope ratios and also immigrants from a distance outside the region of interest by dissimilarities in the multi-isotope fingerprints (Grupe et al., 2018, 2020).

9.3.2 Example 9.2: Birds from Haithabu and Schleswig

Depending on the data set as well as the research question(s), it might be useful or – in the opposite – misleading to remove multivariate outliers prior to the clustering process. Sometimes, mathematical outliers group into a clearly separable cluster. Thus, these individuals could be some of the most important of the whole data set. However, if outliers do not group into a specific cluster, they could also disturb the overall clustering process. In this case, it might be advisable to remove them. An example of a positive effect of outlier removal is demonstrated in this example. Again, stable isotopes had been measured in the bone.

GMM clustering of the complete bird data set including statistical outliers resulted in only two clusters. Cluster 1 contains the majority of individuals ($n = 112$; Fig. 9.2, Table 9.2) and includes individuals with marine or brackish water isotopic signals (Table 9.3). Individuals in cluster 2 are enriched in $^{18}O_{carbonate}$ but depleted in

$^{13}C_{collagen}$, $^{15}N_{collagen}$, and $^{13}C_{carbonate}$ (see Fig. 9.2, Table 9.3). It therefore mainly included birds from terrestrial and freshwater environments (Table 9.2), but still, individuals of the same species were assigned to both clusters. With regard to a pretty high clustering uncertainty for several individuals (Table 9.2) and the overall high standard deviations for all isotope ratios in cluster 1 (Table 9.3), it becomes clear that a higher number of clusters are required to solve the research question.

Seventeen multivariate outliers (see Table 9.2) were detected in the bird data set using the R package "mvoutlier" (Filzmoser & Gschwandtner, 2018). All these individuals were grouped into cluster 1 before (see Table 9.2) but obviously do not form a distinct group (see Fig. 9.2). Therefore, these individuals were removed from the

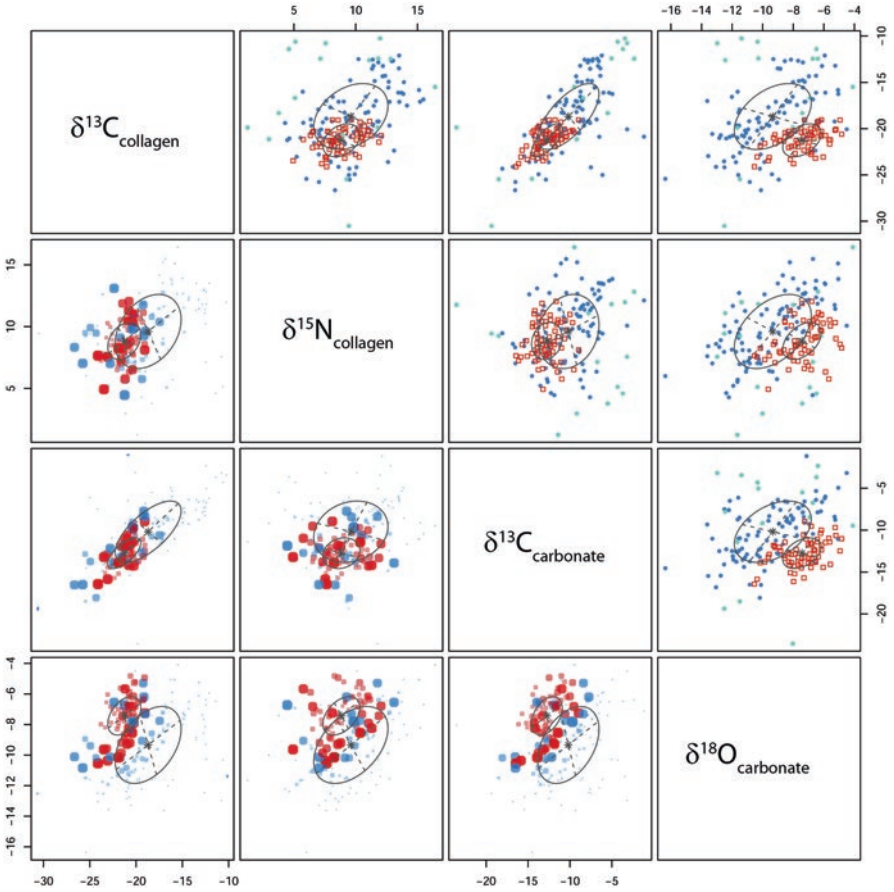


Fig. 9.2 Combined bivariate scatter plots for the four-dimensional bird data set from Haithabu and Schleswig including multivariate outliers indicating classified GMM clusters (above diagonal; cluster 1: blue dots; cluster 2: red squares; multivariate outliers highlighted in turquoise) and uncertainty of the cluster result (below diagonal; cluster 1: blue; cluster 2: red; small symbols: uncertainty <0.1; medium symbols: uncertainty = [0.1, 0.25]; large symbols: uncertainty ≥0.25; color intensity increases with increasing uncertainty). GMM cluster analysis was performed using the "mclust" package (see Scrucca et al., 2016). See also Table 9.2

Table 9.2 GMM clustering results indicated by clustering probability p for the bird data set (sorted by bird species) from Haithabu (H) and Schleswig (S) investigating four-dimensional isotopic data (1) including multivariate outliers and (2) excluding multivariate outliers ("–")

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Black grouse S 97	−21.05	4.63	−12.27	−12.38	**0.999**	0.001	0.000	0.000	0.000	**1.000**
Black grouse 113 BH1H	−21.19	5.76	−10.97	−5.67	0.370	**0.630**	**0.990**	0.000	0.000	0.010
Black grouse 114 BH2H	−20.75	6.34	−11.47	−5.89	0.162	**0.838**	**0.994**	0.000	0.000	0.006
Black grouse 117 BH5Cc	−21.24	4.46	−11.86	−6.73	**0.576**	0.424	**0.948**	0.000	0.000	0.052
Carrion crow S 107	−19.73	10.70	−13.71	−6.53	0.125	**0.875**	**0.988**	0.000	0.004	0.008
Carrion crow S 108	−19.23	11.76	−8.39	−5.29	**0.690**	0.310	**0.894**	0.000	0.077	0.029
Carrion crow S 109	−20.87	9.57	−6.50	−12.15	**1.000**	0.000	0.000	**0.947**	0.000	0.053
Common crane S 101	−20.92	10.51	−10.56	−7.33	0.283	**0.717**	**0.848**	0.000	0.002	0.150
Common crane S 102	−25.00	9.45	−6.12	−11.59	**1.000**	0.000	0.000	**1.000**	0.000	0.000
Common crane 127 KN1Tt	−20.07	9.03	−10.64	−5.21	0.072	**0.928**	**0.998**	0.000	0.000	0.002
Common crane 128 KN2Tt	−21.22	10.06	−14.08	−5.57	0.051	**0.949**	**0.999**	0.000	0.000	0.001
Common crane 129 KN3Tt	−24.17	10.06	−7.49	−12.50	**1.000**	0.000	0.000	**1.000**	0.000	0.000
Common crane 130 KN4U	−19.05	9.64	−11.78	−6.16	0.072	**0.928**	**0.992**	0.000	0.003	0.005
Common crane 131 KN5Cr	−19.30	10.36	−11.09	−6.42	0.093	**0.907**	**0.981**	0.000	0.008	0.011
Common eider Hb 146	−16.57	8.14	−12.48	−12.01	**1.000**	0.000	0.000	0.000	0.010	**0.990**
Common eider Hb 147	−17.10	8.62	−11.23	−9.14	**0.972**	0.028	0.104	0.000	0.239	**0.657**
Common eider 79 EE3Tt	−17.08	9.03	−6.64	−8.38	**0.997**	0.003	0.016	0.002	0.207	**0.775**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Common goldeneye Hb 142	−23.79	7.25	−16.13	−7.98	0.056	**0.944**	**0.940**	0.000	0.000	0.060
Common goldeneye S 61	−22.34	8.54	−15.70	−7.30	0.051	**0.949**	**0.988**	0.000	0.000	0.012
Common goldeneye S 62	−19.06	8.11	−8.96	−7.25	0.405	**0.595**	**0.841**	0.000	0.004	0.155
Common goldeneye S 63	−22.69	8.49	−12.86	−7.88	0.048	**0.952**	**0.897**	0.000	0.000	0.103
Common goldeneye S 65	−18.91	7.63	−12.93	−10.36	**0.969**	0.031	0.028	0.000	0.002	**0.970**
Common goldeneye 137 SC1H	−18.31	10.78	−10.75	−8.58	**0.878**	0.122	0.191	0.000	0.347	**0.462**
Common goldeneye 138 SC2H	−21.61	7.07	−12.06	−9.19	0.236	**0.764**	0.286	0.000	0.000	**0.714**
Common goldeneye 139 SC3H	−26.65	8.33	−16.53	−10.11	**0.667**	0.333	0.007	0.000	0.000	**0.993**
Common goldeneye 140 SC4H	−21.96	8.11	−13.53	−11.31	**0.952**	0.048	0.001	0.000	0.000	**0.999**
Common goldeneye 141 SC5H	−18.81	9.19	−10.18	−10.27	**0.993**	0.007	0.004	0.000	0.006	**0.990**
Common merganser S 76	−25.50	9.51	−12.86	−8.95	**0.894**	0.106	0.012	0.023	0.000	**0.965**
Common merganser S 77	−12.67	13.03	−7.32	−7.68	**1.000**	0.000	0.000	0.000	**1.000**	0.000
Common merganser S 78	−13.49	13.50	−6.37	−9.01	**1.000**	0.000	0.000	0.000	**0.999**	0.001
Common merganser S 79	−20.75	12.05	−11.64	−6.79	0.337	**0.663**	**0.938**	0.000	0.009	0.053
Common merganser S 80	−20.20	7.05	−10.65	−10.35	**0.949**	0.051	0.006	0.000	0.000	**0.994**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Common merganser 120 GS3Tt	−17.09	15.21	−9.14	−8.22	**1.000**	0.000	0.000	0.000	**0.990**	0.010
Common merganser 121 GS4U	−14.98	13.90	−8.58	−8.01	**1.000**	0.000	0.000	0.000	**0.999**	0.001
Common raven Hb 183	−20.20	11.28	−13.26	−11.15	**1.000**	0.000	0.000	0.000	0.002	**0.997**
Common raven Hb 184	−21.07	10.54	−12.30	−6.65	0.058	**0.942**	**0.988**	0.000	0.000	0.012
Common raven Hb 185	−19.66	12.73	−12.45	−11.54	**1.000**	0.000	0.000	0.000	0.006	**0.994**
Common raven KHb 80	−16.12	13.72	−5.32	−6.51	**1.000**	0.000	0.000	0.000	**0.995**	0.004
Common raven 81 KRA2H	−21.67	10.56	−11.42	−5.62	0.097	**0.903**	**0.995**	0.000	0.000	0.005
Common raven 82 KRA3H	−20.83	9.74	−10.38	−6.21	0.103	**0.897**	**0.985**	0.000	0.000	0.015
Common raven 83 KRA4H	−20.67	10.94	−12.37	−6.16	0.064	**0.936**	**0.994**	0.000	0.001	0.005
Cormorant Hb 118	−17.45	10.38	−10.62	−7.76	**0.730**	0.270	0.308	0.000	**0.576**	0.116
Cormorant Hb 119	−13.21	11.44	−8.76	−5.25	**0.998**	0.002	0.000	0.000	**1.000**	0.000
Cormorant Hb 120	−10.25	11.98	−3.43	−11.38	**1.000**	0.000	–	–	–	–
Cormorant Hb 121	−12.85	11.83	−7.87	−7.37	**1.000**	0.000	0.000	0.000	**1.000**	0.000
Cormorant Hb 122	−12.54	11.38	−8.25	−8.15	**1.000**	0.000	0.000	0.000	**1.000**	0.000
Cormorant KHb 88	−17.00	11.63	−8.59	−8.42	**0.996**	0.004	0.004	0.000	**0.887**	0.108
Cormorant 86 K2H	−14.85	11.53	−7.37	−7.63	**0.999**	0.001	0.000	0.000	**0.997**	0.003
Cormorant 87 K3F	−14.29	12.38	−6.37	−6.51	**0.999**	0.001	0.000	0.000	**1.000**	0.000
Cormorant 89 K5U	−14.34	12.18	−9.26	−6.97	**0.999**	0.001	0.000	0.000	**1.000**	0.000

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Domestic chicken Hb 113	−20.54	10.08	−12.91	−8.13	0.146	**0.854**	**0.880**	0.000	0.002	0.118
Domestic chicken Hb 114	−20.41	7.84	−14.51	−7.47	0.115	**0.885**	**0.980**	0.000	0.000	0.020
Domestic chicken Hb 115	−20.21	11.04	−14.27	−7.92	0.353	**0.647**	**0.895**	0.000	0.010	0.095
Domestic chicken Hb 117	−21.49	8.01	−13.34	−6.97	0.027	**0.973**	**0.991**	0.000	0.000	0.009
Domestic chicken S 36	−20.84	7.12	−14.38	−11.22	**0.962**	0.038	0.005	0.000	0.000	**0.995**
Domestic chicken S 37	−20.99	8.56	−11.82	−9.29	0.388	**0.612**	0.220	0.000	0.000	**0.779**
Domestic chicken S 38	−20.46	9.15	−12.45	−6.58	0.033	**0.967**	**0.994**	0.000	0.000	0.006
Domestic chicken S 39	−21.67	9.12	−13.22	−10.18	**0.713**	0.287	0.037	0.000	0.000	**0.963**
Domestic chicken S 40	−21.15	7.12	−11.07	−7.15	0.087	**0.913**	**0.956**	0.000	0.000	0.044
Domestic goose Hb 108	−21.92	8.04	−13.88	−8.12	0.040	**0.960**	**0.936**	0.000	0.000	0.064
Domestic goose Hb 109	−22.24	7.88	−13.83	−6.81	0.026	**0.974**	**0.993**	0.000	0.000	0.007
Domestic goose Hb 110	−23.06	7.54	−15.85	−10.36	0.430	**0.570**	0.072	0.000	0.000	**0.928**
Domestic goose Hb 111	−23.07	10.28	−14.43	−6.04	0.055	**0.945**	**0.996**	0.000	0.000	0.004
Domestic goose Hb 112	−21.93	6.83	−11.77	−7.02	0.070	**0.930**	**0.968**	0.000	0.000	0.032
Domestic goose S 31	−22.05	6.51	−13.76	−8.30	0.068	**0.932**	**0.883**	0.000	0.000	0.117
Domestic goose S 32	−22.58	9.40	−13.95	−7.02	0.029	**0.971**	**0.988**	0.000	0.000	0.012

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Domestic goose S 33	−23.20	9.11	−13.83	−6.55	0.033	**0.967**	**0.992**	0.000	0.000	0.008
Domestic goose S 34	−21.95	7.20	−14.11	−7.68	0.040	**0.960**	**0.972**	0.000	0.000	0.028
Domestic goose S 35	−22.79	9.99	−14.47	−6.84	0.040	**0.960**	**0.990**	0.000	0.000	0.010
European widgeon Hb 126	−22.96	8.43	−15.46	−13.56	**1.000**	0.000	0.000	0.000	0.000	**1.000**
European widgeon Hb 127	−23.03	5.22	−13.52	−11.92	**0.981**	0.019	0.000	0.000	0.000	**1.000**
Goshawk Hb 167	−21.89	8.86	−15.19	−7.91	0.059	**0.941**	**0.965**	0.000	0.000	0.035
Goshawk Hb 171	−19.95	8.67	−13.90	−7.50	0.106	**0.894**	**0.977**	0.000	0.000	0.023
Goshawk S 86	−19.23	11.86	−7.87	−6.41	**0.939**	0.061	**0.473**	0.005	0.245	0.278
Goshawk S 87	−20.07	10.08	−3.17	−4.50	**1.000**	0.000	0.000	0.998	0.000	0.002
Goshawk S 88	−20.60	8.28	−11.02	−4.83	0.102	**0.898**	**0.999**	0.000	0.000	0.001
Goshawk S 89	−20.38	6.52	−9.53	−6.82	0.349	**0.651**	**0.918**	0.000	0.000	0.081
Goshawk S 90	−19.16	9.26	−7.74	−6.20	**0.647**	0.353	**0.888**	0.001	0.010	0.102
Goshawk 122 Hb1F	−20.78	11.29	−14.18	−8.24	0.414	**0.586**	**0.805**	0.000	0.007	0.188
Goshawk 123 Hb2F	−19.91	9.18	−11.40	−6.43	0.047	**0.953**	**0.992**	0.000	0.001	0.007
Goshawk 124 Hb3F	−20.27	6.60	−13.26	−7.27	0.128	**0.872**	**0.980**	0.000	0.000	0.020
Goshawk 125 Hb4F	−19.08	8.58	−12.48	−4.94	0.163	**0.837**	**0.998**	0.000	0.000	0.001
Goshawk 126 Hb5F	−19.98	10.18	−11.58	−6.87	0.075	**0.925**	**0.979**	0.000	0.003	0.018
Greater scaup Hb 140	−20.71	8.91	−15.43	−12.74	**1.000**	0.000	0.000	0.000	0.000	**1.000**
Greater scaup S 51	−25.42	6.01	−14.53	−16.37	**1.000**	0.000	–	–	–	–
Greater scaup S 52	−14.78	9.56	−7.88	−7.38	**0.994**	0.006	0.003	0.000	**0.987**	0.010
Greater scaup S 54	−21.25	7.98	−12.07	−8.91	0.175	**0.825**	0.496	0.000	0.000	**0.504**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Guillemot Hb 178	−17.50	14.15	−11.68	−8.68	**1.000**	0.000	0.000	0.000	**0.969**	0.031
Guillemot Hb 179	−17.10	12.74	−10.43	−7.21	**0.959**	0.041	0.013	0.000	**0.978**	0.010
Guillemot Hb 180	−15.83	15.12	−11.06	−5.07	**0.998**	0.002	0.000	0.000	**1.000**	0.000
Guillemot Hb 182	−15.50	16.43	−9.47	−4.10	**1.000**	0.000	–	–	–	–
Guillemot 109 TL2St	−17.11	13.53	−6.57	−7.26	**1.000**	0.000	0.000	0.001	**0.970**	0.028
Guillemot 110 TL3St	−16.25	13.86	−4.94	−5.05	**1.000**	0.000	0.000	0.001	**0.998**	0.002
Mallard Hb 131	−24.11	7.65	−16.44	−10.54	0.474	**0.526**	0.031	0.000	0.000	**0.969**
Mallard Hb 133	−21.26	8.42	−12.29	−7.81	0.050	**0.950**	**0.937**	0.000	0.000	0.063
Mallard Hb 134	−21.29	7.86	−12.45	−8.83	0.130	**0.870**	**0.617**	0.000	0.000	0.383
Mallard S 46	−20.60	10.41	−12.85	−6.91	0.058	**0.942**	**0.987**	0.000	0.001	0.013
Mallard S 47	−21.47	11.88	−13.32	−7.43	0.294	**0.706**	**0.918**	0.000	0.003	0.079
Mallard S 48	−23.33	7.60	−15.09	−8.19	0.039	**0.961**	**0.917**	0.000	0.000	0.083
Mallard S 49	−20.43	10.36	−11.44	−8.54	0.466	**0.534**	0.472	0.000	0.007	**0.521**
Mallard S 50	−23.60	12.62	−14.85	−9.88	**0.996**	0.004	0.005	0.000	0.000	**0.994**
Mallard 104 SE2H	−23.13	7.64	−12.89	−9.57	0.282	**0.718**	0.089	0.000	0.000	**0.911**
Mallard 105 SE3U	−20.35	9.84	−9.71	−7.82	**0.503**	0.497	**0.567**	0.000	0.004	0.429
Mallard 152 SE1H	−16.84	15.48	−10.45	−6.71	**1.000**	0.000	0.000	0.000	**0.999**	0.001
Mallard 153 SE2H	−12.71	12.92	−6.56	−8.55	**1.000**	0.000	0.000	0.000	**1.000**	0.000
Mallard 154 SE3H	−21.54	8.56	−13.23	−7.69	0.034	**0.966**	**0.967**	0.000	0.000	0.033
Mallard 155 SE4H	−19.63	10.43	−13.77	−11.72	**1.000**	0.000	0.000	0.000	0.001	**0.999**
Mallard 156 SE5H	−23.47	4.93	−13.21	−9.63	0.442	**0.558**	0.049	0.000	0.000	**0.951**
Mute swan Hb 159	−11.41	4.60	−3.68	−6.48	**1.000**	0.000	–	–	–	–

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Mute swan Hb 160	−20.86	12.13	−1.16	−7.13	**1.000**	0.000	0.000	**1.000**	0.000	0.000
Mute swan Hb 161	−10.61	5.13	−4.30	−10.33	**1.000**	0.000	–	–	–	–
Peregrine falcon S 91	−20.29	10.99	−10.00	−6.16	0.242	**0.758**	**0.970**	0.000	0.004	0.026
Peregrine falcon S 92	−20.48	10.84	−8.99	−8.38	**0.968**	0.032	0.050	0.008	0.006	**0.936**
Peregrine falcon S 93	−19.01	9.66	−8.42	−7.31	**0.681**	0.319	**0.628**	0.000	0.038	0.334
Peregrine falcon S 94	−19.20	8.08	−8.84	−10.88	**1.000**	0.000	0.000	0.001	0.000	**0.999**
Pintail duck Hb 123	−12.45	7.14	−6.73	−6.72	**1.000**	0.000	–	–	–	–
Pintail duck Hb 124	−19.36	5.76	−12.73	−9.06	**0.746**	0.254	0.423	0.000	0.000	**0.577**
Pintail duck Hb 125	−24.32	9.48	−18.06	−10.18	**0.796**	0.204	0.059	0.000	0.000	**0.941**
Pintail duck S 56	−20.94	10.44	−13.38	−8.55	0.266	**0.734**	**0.726**	0.000	0.002	0.271
Pintail duck S 57	−23.52	7.96	−14.08	−12.52	**0.998**	0.002	0.000	0.000	0.000	**1.000**
Pintail duck S 58	−12.49	10.23	−7.25	−9.30	**1.000**	0.000	0.000	0.000	**0.997**	0.003
Red breasted merganser S 71	−12.60	11.09	−7.78	−12.45	**1.000**	0.000	–	–	–	–
Red breasted merganser S 72	−20.47	6.36	−11.17	−11.09	**0.987**	0.013	0.001	0.000	0.000	**0.999**
Red breasted merganser S 73	−20.67	8.78	−11.92	−9.20	0.377	**0.623**	0.296	0.000	0.001	**0.703**
Red breasted merganser S 74	−13.53	11.33	−7.58	−8.84	**1.000**	0.000	0.000	0.000	**0.998**	0.002
Red breasted merganser S 75	−14.93	12.67	−9.11	−7.25	**0.999**	0.001	0.000	0.000	**0.999**	0.001
Shelduck Hb 148	−20.75	11.68	−13.10	−11.95	**1.000**	0.000	0.000	0.000	0.000	**1.000**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Shelduck Hb 149	−17.01	7.61	−9.31	−13.62	**1.000**	0.000	0.000	0.000	0.000	**1.000**
Shelduck Hb 150	−19.21	11.13	−13.16	−6.36	0.176	**0.824**	**0.972**	0.000	0.019	0.009
Smew S 66	−18.26	9.10	−10.98	−11.57	**1.000**	0.000	0.000	0.000	0.003	**0.997**
Tufted duck Hb 136	−25.46	9.26	−18.49	−11.49	**0.983**	0.017	–	–	–	–
Tufted duck Hb 138	−21.03	7.49	−13.73	−10.83	**0.874**	0.126	0.013	0.000	0.000	**0.987**
Tufted duck Hb 139	−21.36	8.12	−13.06	−8.61	0.083	**0.917**	**0.796**	0.000	0.000	0.204
Tufted duck 90 RE1H	−22.79	7.02	−12.19	−10.13	**0.639**	0.361	0.012	0.000	0.000	**0.988**
Tufted duck 91 RE2Tt	−10.77	7.57	−3.18	−12.97	**1.000**	0.000	–	–	–	–
Tufted duck 92 RE3St	−22.61	9.82	−10.95	−11.49	**1.000**	0.000	0.000	0.018	0.000	**0.982**
Tufted duck 132 RE1H	−19.31	7.34	−12.32	−10.24	**0.924**	0.076	0.035	0.000	0.001	**0.964**
Tufted duck 133 RE2H	−24.12	8.61	−16.81	−13.05	**1.000**	0.000	0.000	0.000	0.000	**1.000**
Tufted duck 134 RE3H	−25.76	7.03	−16.50	−10.84	**0.623**	0.377	0.003	0.000	0.000	**0.997**
Tufted duck 135 RE4H	−30.51	9.44	−19.36	−12.51	**1.000**	0.000	–	–	–	–
Tufted duck 136 RE5H	−20.29	7.71	−12.75	−8.29	0.110	**0.890**	**0.889**	0.000	0.000	0.111
White-fronted goose Hb 151	−22.56	8.94	−13.41	−6.80	0.026	**0.974**	**0.990**	0.000	0.000	0.010
White-fronted goose Hb 152	−21.00	5.54	−12.64	−11.65	**0.991**	0.009	0.000	0.000	0.000	**1.000**
White-tailed eagle Hb 162	−19.87	11.77	−23.54	−8.03	**1.000**	0.000	–	–	–	–
White-tailed eagle Hb 163	−19.91	10.97	−13.73	−10.07	**0.980**	0.020	0.028	0.000	0.014	**0.958**
White-tailed eagle Hb 164	−24.93	9.92	−13.74	−9.39	**0.786**	0.214	0.021	0.002	0.000	**0.977**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
White-tailed eagle Hb 166	−17.53	11.46	−9.56	−8.71	**0.991**	0.009	0.014	0.000	**0.736**	0.249
White-tailed eagle KHb 94	−18.20	11.02	−9.21	−8.93	**0.989**	0.011	0.022	0.000	0.286	**0.691**
White-tailed eagle KHb 96	−12.10	13.51	−6.20	−5.94	**1.000**	0.000	0.000	0.000	**1.000**	0.000
White-tailed eagle KHb 97	−16.11	13.45	−6.00	−5.90	**0.999**	0.001	0.000	0.000	**0.998**	0.002
White-tailed eagle KS 144	−12.45	12.83	−2.33	−6.39	**1.000**	0.000	–	–	–	–
White-tailed eagle KS 146	−13.24	12.20	−6.64	−6.12	**1.000**	0.000	0.000	0.000	**1.000**	0.000
White-tailed eagle S 81	−20.18	10.32	−9.68	−5.26	0.177	**0.823**	**0.993**	0.000	0.001	0.006
White-tailed eagle S 82	−16.55	10.83	−8.74	−8.04	**0.981**	0.019	0.013	0.000	**0.925**	0.063
White-tailed eagle S 83	−12.41	11.83	−5.03	−10.28	**1.000**	0.000	–	–	–	–
White-tailed eagle S 84	−20.47	9.87	−8.89	−8.45	**0.929**	0.071	0.066	0.005	0.002	**0.926**
White-tailed eagle S 85	−15.56	12.64	−7.06	−7.37	**0.999**	0.001	0.000	0.000	**0.997**	0.003
White-tailed eagle 93 SA1U	−19.88	1.25	−11.41	−11.65	**1.000**	0.000	–	–	–	–
White-tailed eagle 95 SA3U	−19.54	11.41	−11.24	−9.73	**0.991**	0.009	0.012	0.000	0.034	**0.954**
White-tailed eagle 142 SA1H	−22.37	13.10	−13.92	−6.55	**0.625**	0.375	**0.952**	0.000	0.001	0.046
Whooper swan Hb 153	−17.69	7.95	−8.87	−9.62	**0.993**	0.007	0.011	0.000	0.019	**0.970**

(continued)

Table 9.2 (continued)

Individual	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	Including outliers		Without outliers			
					p1	p2	p1	p2	p3	p4
Whooper swan Hb 154	−21.90	8.25	−13.89	−10.15	0.484	**0.516**	0.075	0.000	0.000	**0.925**
Whooper swan Hb 155	−21.60	7.33	−13.78	−9.86	0.330	**0.670**	0.181	0.000	0.000	**0.819**
Whooper swan Hb 156	−19.00	6.36	−6.82	−9.36	**0.999**	0.001	0.001	0.011	0.000	**0.988**
Whooper swan S 41	−19.68	6.42	−8.32	−9.57	**0.990**	0.010	0.004	0.002	0.000	**0.994**
Whooper swan S 42	−19.39	4.29	−7.60	−12.21	**1.000**	0.000	0.000	0.016	0.000	**0.984**
Whooper swan S 43	−23.62	7.97	−12.53	−11.75	**0.996**	0.004	0.000	0.002	0.000	**0.998**
Whooper swan S 44	−21.55	6.64	−12.88	−11.03	**0.903**	0.097	0.003	0.000	0.000	**0.997**
Whooper swan S 45	−16.01	5.71	−8.73	−11.63	**1.000**	0.000	0.000	0.000	0.001	**0.999**
Whooper swan 98 SW1H	−20.08	7.21	−10.58	−10.95	**0.992**	0.008	0.001	0.000	0.000	**0.999**
Whooper swan 99 SW2Cc	−18.32	3.78	−5.50	−7.41	**1.000**	0.000	–	–	–	–
Whooper swan 100 SW3R	−16.69	5.08	−8.44	−6.92	**0.991**	0.009	–	–	–	–
Whooper swan 147 SW1R	−20.57	7.03	−11.02	−13.29	**1.000**	0.000	0.000	0.001	0.000	**0.999**
Whooper swan 148 SW2R	−21.14	2.98	−9.17	−10.01	**0.999**	0.001	–	–	–	–
Whooper swan 149 SW3H	−17.94	6.97	−8.50	−11.51	**1.000**	0.000	0.000	0.000	0.000	**0.999**
Whooper swan 150 SW4F	−20.73	8.68	−12.19	−11.35	**0.993**	0.007	0.001	0.000	0.000	**0.999**
Whooper swan 151 SW5Tt	−21.01	8.36	−12.49	−12.84	**1.000**	0.000	0.000	0.000	0.000	**1.000**

Clustering probabilities for allocated clusters are printed in bold

Table 9.3 Mean (μ) and standard deviation (sd) of the bird data set from Haithabu and Schleswig including multivariate outliers

μ	**Cluster 1**	**Cluster 2**
$\delta^{13}C_{collagen}$	−18.69	−21.27
$\delta^{15}N_{collagen}$	9.61	8.84
$\delta^{13}C_{carbonate}$	−10.19	−12.75
$\delta^{18}O_{carbonate}$	−9.34	−7.43
sd	**Cluster 1**	**Cluster 2**
$\delta^{13}C_{collagen}$	3.56	1.78
$\delta^{15}N_{collagen}$	3.00	1.50
$\delta^{13}C_{carbonate}$	3.66	1.83
$\delta^{18}O_{carbonate}$	2.52	1.26

data set. After outlier removal, an optimal number of four clusters are detected in the data set (n = 160; Table 9.2, Fig. 9.3). Only a handful of individuals could not be assigned to one of the four clusters with a sufficient certainty and were captured by two clusters with a roughly 50:50 probability each (e.g., greater scaup S 54; see Table 9.2). These individuals plot on the border of two or even three cluster ellipses (see Fig. 9.3).

Birds found in cluster 1 (n = 61) are enriched in $^{18}O_{carbonate}$ (Table 9.4). With the exception of only seven individuals, they had been grouped into cluster 2 ("with outlier experiment") before (Table 9.2). This cluster now mainly includes terrestrial or freshwater birds, including the majority of domesticated chicken and geese. Cluster 2 is particularly small and includes only five individuals (Table 9.2) that differ from the other birds mainly with respect to $\delta^{13}C_{carbonate}$ ($\mu = -4.98‰$; Table 9.4). These individuals were probably primarily not local to the sites because they constitute isotopic rather than mathematical outliers.

Birds grouped into cluster 3 (n = 30; Table 9.2) are enriched in all isotopic systems (see Table 9.4). The majority of these individuals are piscivores, e.g., cormorant, guillemot, and white-tailed eagle. According to the archaeological evidence, guillemots were imported from their breeding grounds on Heligoland (Reichstein, 1984; Requate, 1956); however, the overall strong marine isotope signal in individuals of cluster 3 does not permit such a distinction. It needs to be emphasized that the $\delta^{18}O_{carbonate}$ values of the birds in cluster 1 ($\mu = -7.03‰$) are similar to those of the birds in cluster 3 ($\mu = -7.38‰$; Fig. 9.3, Table 9.4). This indicates a marine impact on the birds in cluster 1, what cannot be explained by the individuals' dietary preferences (see Fig. 9.3, Table 9.4). The most plausible explanation is the marked influence of the sea spray effect on $\delta^{18}O_{carbonate}$ (see Göhring et al., 2018).

Cluster 4 finally captured individuals (n = 64) that were depleted in all isotopic dimensions (Fig. 9.3, Table 9.4), indicating a freshwater or terrestrial signal. Several bird species found in this cluster were not local to the Baltic Sea but winter visitors that time, namely, European widgeon, greater scaup, smew, tufted duck, and whooper swan (see Reichstein, 1974, 1984; Reichstein & Pieper, 1986). With the exception of one greater scaup (S 52) and two tufted ducks (Hb 139, 136 RE5H),

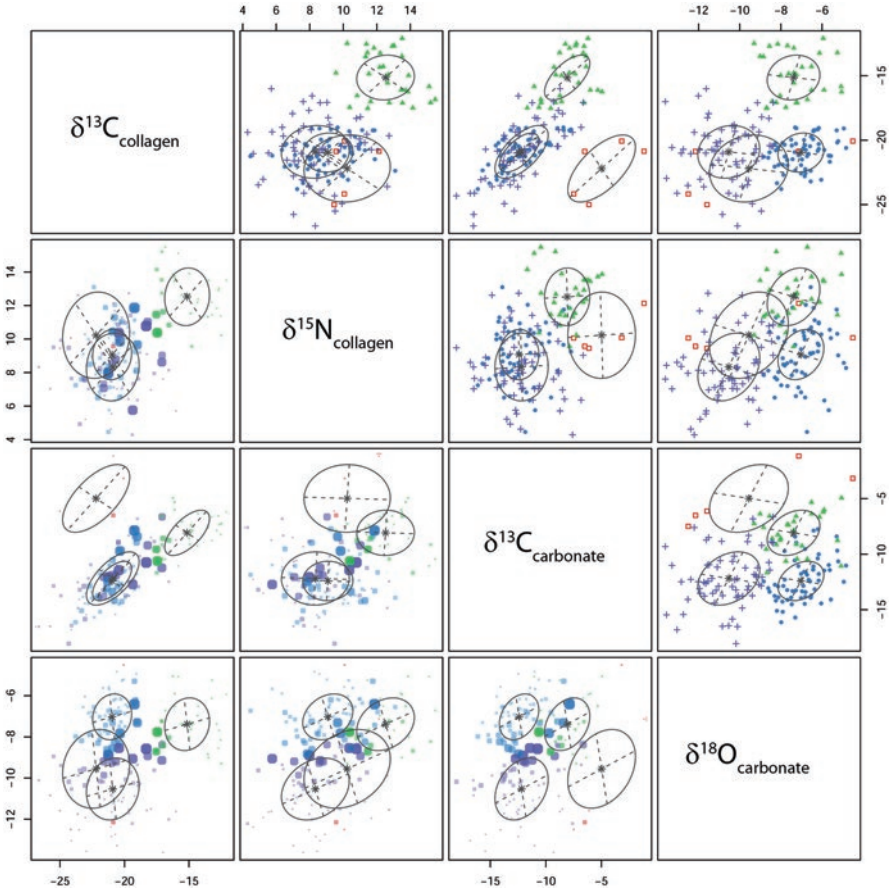


Fig. 9.3 Bivariate scatter plots for the bird data set from Haithabu and Schleswig after removal of multivariate outliers indicating classified GMM clusters (above diagonal; cluster 1: blue dots; cluster 2: red squares; cluster 3: green triangles; cluster 4: purple plusses) and uncertainty of the cluster result (below diagonal; cluster 1: blue; cluster 2: red; cluster 3: green; cluster 4: purple; small symbols: uncertainty <0.1; medium symbols: uncertainty = [0.1, 0.25]; large symbols: uncertainty ≥0.25; color intensity increases with increasing uncertainty). GMM cluster analysis was performed using the "mclust" package (see Scrucca et al., 2016). See also Table 9.2

winter visitors were restricted to cluster 4 (Table 9.2). Greater scaups are flexible and live in both freshwater (Hb 140, S 54) and marine (S 52) habitats. With regard to the two tufted ducks, a sea spray signal is again evidenced (cluster 1, see the lower clustering probability p in Table 9.2). All other winter visitors exhibit a (very) high probability for cluster 4 (see Table 9.2) and lack any marine impact in their isotopic signatures (see Table 9.4).

Besides these winter visitors, some individuals of the same species are nevertheless captured by several clusters (carrion crow, common crane, common goldeneye, common raven, domestic chicken, domestic goose, goshawk, greater scaup,

Table 9.4 Mean (μ) and standard deviation (sd) of the bird data set from Haithabu and Schleswig after removal of multivariate outliers

μ	Cluster 1	Cluster 2	Cluster 3	Cluster 4
$\delta^{13}C_{collagen}$	−20.96	−22.20	−15.14	−20.91
$\delta^{15}N_{collagen}$	9.08	10.23	12.52	8.33
$\delta^{13}C_{carbonate}$	−12.41	−4.98	−8.08	−12.19
$\delta^{18}O_{carbonate}$	−7.03	−9.55	−7.38	−10.53
sd	**Cluster 1**	**Cluster 2**	**Cluster 3**	**Cluster 4**
$\delta^{13}C_{collagen}$	1.51	1.50	1.78	1.11
$\delta^{15}N_{collagen}$	2.60	2.58	3.05	1.92
$\delta^{13}C_{carbonate}$	1.74	1.72	2.04	1.28
$\delta^{18}O_{carbonate}$	2.04	2.02	2.40	1.51

mallard, peregrine falcon, pintail duck, red-breasted merganser, shelduck, white-fronted goose, white-tailed eagle; Table 9.2). All these species have in common that they are very flexible with respect to their habitat choice and/or hunting grounds (see Reichstein, 1974, 1984; Reichstein & Pieper, 1986). White-tailed eagles, for example, obviously hunted both terrestrial or freshwater prey (cluster 3) and marine prey (cluster 4) but are also partly influenced by sea spray (cluster 1).

The present example illustrates that GMM cluster analysis is able to differentiate between different diets and habitats, even in a very complex environment. By the use of GMM cluster analysis, both habitat and hunting grounds of the species were identified which is of particular importance in palaeoecology (e.g., changing environmental conditions over time).

Comparison of the GMM result after outlier removal with the conventional bivariate plots of isotope ratios clearly demonstrates the superiority of the former. In the pairwise bivariate plots (Fig. 9.4), individual birds with more positive $\delta^{13}C_{collagen}$ and $\delta^{15}N_{collagen}$ values (those that had been grouped into cluster 3 in Fig. 9.3) are easily identified. Unsurprisingly, this is the group of marine fish consumers, for example, cormorants and white-tailed eagles. Even labeling the species does not provide more structure in this data set. Only cluster analysis was capable of detecting clusters 1, 2, and 4 according to the similarities of the four-dimensional isotopic fingerprint. This example demonstrates that a grouping according to palaeoecological parameters by GMM analysis is possible, even in cases where no *a priori* such knowledge would be available.

GMM can be compared with a Principle Component Analysis (PCA), a method frequently applied to multidimensional data. PCA results for the bird data set excluding multivariate outliers are shown in Fig. 9.5. The first component (PC 1; 56.3%) is mainly represented by $\delta^{13}C_{collagen}$, $\delta^{15}N_{collagen}$, and $\delta^{13}C_{carbonate}$, while the second component (PC 2; 22.1%) is largely represented by $\delta^{18}O_{carbonate}$ (Table 9.5). Without any labeling, only two groups are detected by PCA, whereby one of these groups includes birds that had been grouped into cluster 3 and additional individuals of other clusters (see Fig. 9.5). After adding the GMM cluster labels, PCA is able to graphically separate three of the four GMM clusters, while individuals of cluster 2 are not captured by PCA at all (Fig. 9.5). Moreover, PCA requires *a priori*

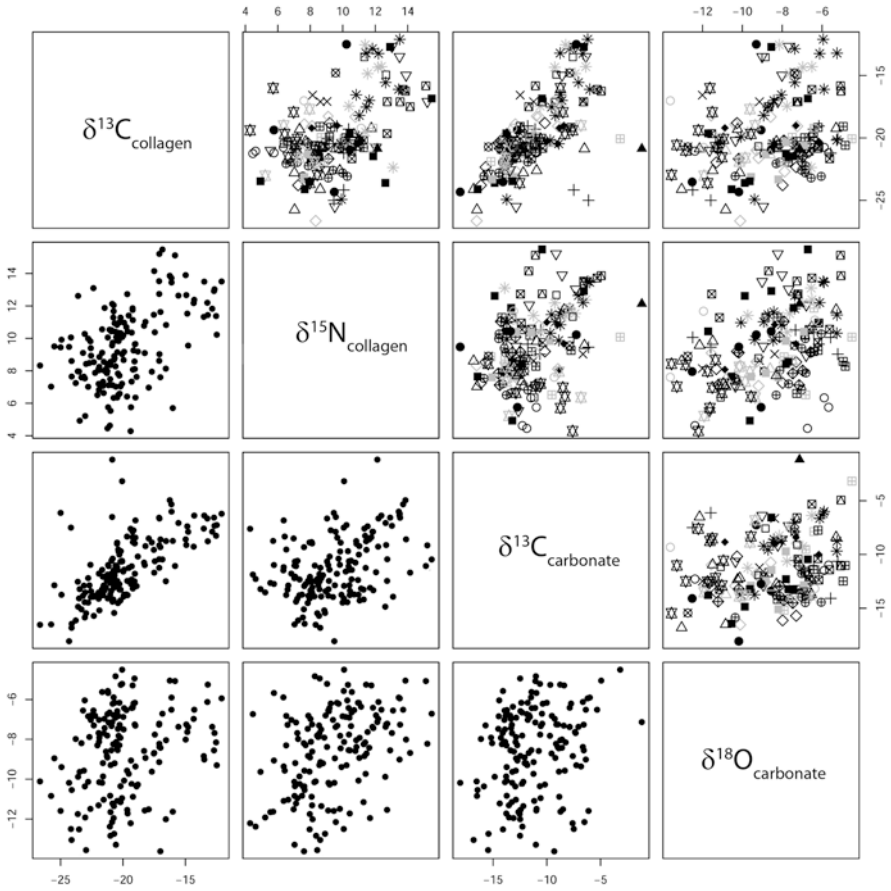


Fig. 9.4 Bivariate scatter plots for the bird data set from Haithabu and Schleswig after removal of multivariate outliers with (above diagonal) and without species labeling (below diagonal)

knowledge of group membership what is usually not known beforehand. Since neither habitat preference nor feeding habit permits a direct detection of groups in the bird data set; no labels other than clustering labels can be applied to the data. In this example, PCA is no more than a possible way of illustrating already detected GMM cluster labels rather than a method for the detection of groups within multi-isotope data according to their similarities (see also Mauder et al., 2019).

9.3.3 Example 9.3: Humans from Haithabu and Schleswig

The third example shall illustrate the benefit obtained by combining GMM clustering results with another method of interpreting stable isotope data that has become more common in recent times, namely, isotopic sourcing by the use of stable isotope

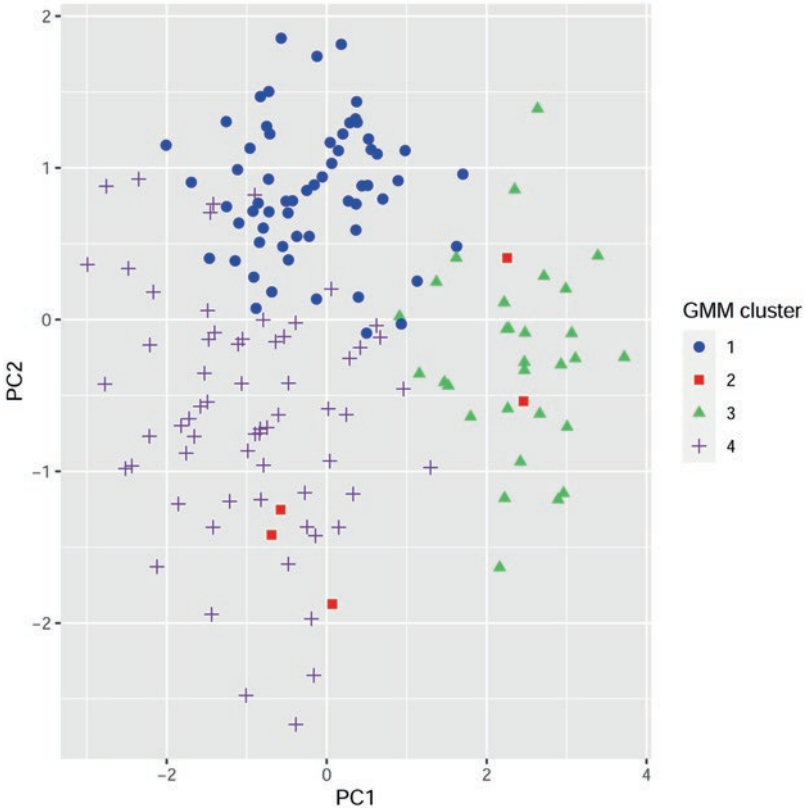


Fig. 9.5 Biplot of the first two principal components for the bird data set from Haithabu and Schleswig after removal of multivariate outliers. The allocated GMM cluster labels are illustrated by different colors (see Table 9.2). Cumulated variance: PC 1, 0.563; PC 2, 0.784; PC 3, 0.924; and PC 4, 1.000

Table 9.5 Loading values for principal component analysis of the bird data set from Haithabu and Schleswig after removal of multivariate outliers

Loading	PC1	PC2	PC3	PC4
$\delta^{13}C_{collagen}$	0.57	−0.35	−0.02	0.75
$\delta^{15}N_{collagen}$	0.51	0.25	0.78	−0.25
$\delta^{13}C_{carbonate}$	0.53	−0.43	−0.40	−0.61
$\delta^{18}O_{carbonate}$	0.37	0.79	−0.47	0.08

mixing models (here: SISUS; Erhardt, 2014). A six-dimensional ($\delta^{13}C_{collagen}$, $\delta^{15}N_{collagen}$, $\delta^{13}C_{carbonate}$, $\delta^{18}O_{carbonate}$, $\delta^{34}S_{collagen}$, and $^{87}Sr/^{86}Sr$) isotopic fingerprint has been established for 36 adult human skeletons from the Haithabu and Schleswig sites (Table 9.6). $^{87}Sr/^{86}Sr$ had been measured in the enamel of the first permanent molar that reflects the place of childhood residence. Stable isotope ratios of the light elements were measured in bone collagen or structural carbonate of the same

individuals. The SISUS model had been applied to the diet-adjusted $\delta^{13}C$ and $\delta^{15}N$ values. Chosen end-members were the pooled data of 27 domesticated mammals (cattle, pig, sheep), seven freshwater fish (bream, pike, tench), eight marine fish (cod, haddock, perch, plaice, sturgeon), and 25 marine mammals (common seal, gray seal, ringed seal, saddleback seal) (for original data, see Doppler et al., 2010). The source contribution by plants was calculated according to Krueger and Sullivan (1984). Carbon and nitrogen concentrations for humans and animals were taken from Doppler et al. (2010). Plant carbon and nitrogen concentration was chosen according to Hagl et al. (2013) (for the SISUS results, see Göhring et al., 2020c).

The optimal clustering model results in three clusters with an average clustering probability of 0.973 (see Table 9.6). A small group of individuals assigned to cluster 2 (Fig. 9.6) is readily detectable by conspicuous fingerprints. Two individuals with $^{87}Sr/^{86}Sr$ isotope ratios >0.737, for example, are both primarily nonlocal to the sites and probably originated from Northern Scandinavia (see, e.g., Lahtinen et al., 2021; Price et al., 2018, 2019). Interestingly, both are females, suggestive of exogamy. Individuals in cluster 2 also differ from the other individuals with respect to their diet (Fig. 9.7). Based on the SISUS results, protein of both marine mammals and marine fish contributed much more to the regular diet (ca. 45%) compared to the individuals of the other two clusters (ca. 29%), while the consumption of protein from domesticated mammals (cattle, sheep, pig) was lower (ca. 16%) compared to cluster 1 (26%) and cluster 3 (24%) (Göhring et al., 2020c). While a different diet is not necessarily due to a nonlocal origin, the overall isotopic fingerprints of the individuals in cluster 2 make a nonlocal origin plausible. It has to be kept in mind, however, that the sea spray effect can mask the nonlocal origin of some other individuals (Göhring et al., 2020c), including $^{87}Sr/^{86}Sr$ (see, e.g., Alonzi et al., 2020; Göhring et al., 2023; Kusaka et al., 2009; Snoeck, 2014).

Clusters 1 and 3 are almost identical with respect to $\delta^{13}C_{collagen}$, $\delta^{15}N_{collagen}$, and $\delta^{13}C_{carbonate}$; they, however, differ from each other with respect to their $\delta^{18}O_{carbonate}$, $\delta^{34}S_{collagen}$, and $^{87}Sr/^{86}Sr$ values (see Fig. 9.6, Table 9.7). Differences in the individual diets are thus not detectable (see Fig. 9.6, Table 9.7). The SISUS mixing model was not able to calculate the protein sources for five (14%) of the individuals tested, namely, RM 63, RM 103, RM 194, H 270, and H 12301 K I/1 (see Fig. 9.7). It is almost certain that the animal bones chosen for the definition of the end-members are not the leftovers of the food consumed by these human individuals. Therefore, human diet-adjusted stable isotope ratios that plot only slightly outside the mixing polygon do not indicate a substantially different dietary behavior. Only the collagen isotope mixture of the young female RM 194 (cluster 2) cannot be explained by the five food sources chosen for the model. In combination with the overall isotopic fingerprint of this individual, a nonlocal origin is indicated.

Both GMM clustering and isotopic sourcing reveal differences which can explain the observed divergence in $\delta^{18}O_{carbonate}$, $\delta^{34}S_{collagen}$, and $^{87}Sr/^{86}Sr$. The diet of individuals assigned to cluster 3 contained a somewhat higher amount of freshwater fish (up to 47%), compared to the others (cluster 1, up to 27%, cluster 2, up to 23%; see also Fig. 9.7). Accordingly, besides a marine signal (partly caused by sea spray; see

Table 9.6 GMM clustering results indicated by clustering probability p for the data set of at least adult humans from Haithabu and Schleswig

Site	Individual	Sex	Age	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	$\delta^{34}S_{collagen}$ [%]	$^{87}Sr/^{86}Sr$	p1	p2	p3
HB	H 17	m	Adult	−19.88	10.37	−13.37	−4.97	11.89	0.711713	0.002	0.000	**0.998**
HB	H 66a	nd	Adult	−19.70	11.28	−14.92	−5.36	12.02	0.711824	0.001	0.000	**0.999**
HB	H 72	m	Adult	−20.15	11.49	−15.52	−4.93	10.12	0.711807	0.006	0.000	**0.994**
HB	H 105	m	Adult	−20.24	11.42	−11.62	−7.24	7.48	0.711176	**1.000**	0.000	0.000
HB	H 106a	m	Adult	−19.88	12.06	−14.79	−7.18	13.30	0.711698	0.011	0.001	**0.988**
HB	H 270	f	Adult	−19.80	12.86	−15.57	−6.35	10.38	0.712048	0.010	0.001	**0.989**
HB	H 271	m	Adult	−19.58	11.73	−14.50	−7.72	4.42	0.711613	**0.999**	0.001	0.000
HB	H 274	m	Adult	−19.79	9.96	−13.64	−4.83	13.70	0.711795	0.003	0.000	**0.997**
HB	H 275	m	Adult	−19.88	10.35	−14.62	−5.85	12.47	0.711726	0.001	0.000	**0.999**
HB	H 1904 III/1	m	Adult	−19.71	11.01	−13.63	−7.04	11.45	0.711316	0.091	0.000	**0.909**
HB	H 1912 XIII/3	f	Adult	−20.14	12.03	−14.91	−6.03	11.31	0.712084	0.004	0.000	**0.996**
HB	H 1912 XIV.3	f	Adult	−20.23	11.43	−14.18	−6.46	10.29	0.711826	0.012	0.000	**0.987**
HB	H 1936 DIV	nd	Adult	−19.87	10.96	−14.68	−6.09	5.05	0.710986	**1.000**	0.000	0.000
HB	H 12301 K I/1	nd	Adult	−19.21	13.35	−14.99	−6.00	11.31	0.711481	0.010	0.002	**0.988**
HB	H 12370 K IX/6	f	Adult	−19.94	12.43	−13.31	−5.37	11.16	0.711628	0.004	0.000	**0.996**
RM early	RM 103	m	Late adult	−19.49	13.54	−15.29	−6.91	6.79	0.710600	**0.989**	0.011	0.000
RM early	RM 116	f	Adult	−19.41	11.15	−14.70	−7.32	2.10	0.709810	**1.000**	0.000	0.000
RM early	RM 137	m	Early adult	−19.47	12.00	−14.89	−8.46	11.50	0.709250	**0.970**	0.030	0.000
RM early	RM 140	m	Early adult	−19.54	12.20	−13.87	−7.02	9.54	0.709630	**0.997**	0.003	0.000
RM early	RM 158	f	Early adult	−19.65	10.34	−13.98	−7.57	5.84	0.709330	**1.000**	0.000	0.000

(continued)

Table 9.6 (continued)

Site	Individual	Sex	Age	$\delta^{13}C_{collagen}$ [%]	$\delta^{15}N_{collagen}$ [%]	$\delta^{13}C_{carbonate}$ [%]	$\delta^{18}O_{carbonate}$ [%]	$\delta^{34}S_{collagen}$ [%]	$^{87}Sr/^{86}Sr$	p1	p2	p3
RM early	RM 170	m	Late adult	−19.21	11.49	−14.69	−9.21	10.25	0.709560	**0.967**	0.033	0.000
RM early	RM 194	f	18 a	−21.32	12.86	−15.24	−9.30	6.63	0.737470	0.000	**1.000**	0.000
RM early	RM 198	f	18–20 a	−20.00	11.69	−12.92	−8.07	12.25	0.709540	**0.999**	0.001	0.000
RM early	RM 202	f	Late adult	−18.83	12.61	−15.32	−8.88	12.59	0.738570	0.000	**1.000**	0.000
RM early	RM 208	f	Late adult	−19.50	11.72	−14.93	−7.34	11.08	0.711280	0.196	0.002	**0.801**
RM early	RM 232	f	Early adult	−19.71	10.74	−15.57	−8.75	8.12	0.710640	**0.999**	0.001	0.000
RM late	RM 63	m	Late adult	−19.36	12.67	−14.68	−6.84	11.21	0.711220	0.121	0.005	**0.874**
RM late	RM 142a	m	Early adult	−17.65	12.74	−13.92	−8.05	9.34	0.709420	0.000	**1.000**	0.000
RM late	RM 145	f	Late adult	−19.53	10.68	−15.46	−7.09	11.42	0.711600	0.025	0.000	**0.975**
RM late	RM 159	m	Late adult	−19.24	11.32	−14.41	−7.36	10.15	0.709330	**0.988**	0.012	0.000
RM late	RM 197	m	Early adult	−18.91	12.95	−14.10	−8.80	11.05	0.711520	0.281	**0.709**	0.010
SC	SC 5	m	Adult	−20.24	10.81	−13.81	−5.02	7.78	0.709480	**1.000**	0.000	0.000
SC	SC 92	m	Adult	−17.24	14.34	−14.04	−6.30	12.67	0.711570	0.000	**1.000**	0.000
SC	SC 101	m	Adult	−20.43	9.96	−14.60	−9.99	13.15	0.714920	0.000	**1.000**	0.000
SC	SC 105	m	Adult	−19.62	11.13	−13.73	−5.85	14.03	0.711460	0.022	0.000	**0.978**
SC	SC 232	m	Adult	−19.52	10.37	−14.06	−10.65	11.85	0.711590	**0.935**	0.065	0.000

Clustering probabilities for allocated clusters are printed in bold. *HB* Haithabu, 804–1066 AD, *RM* early, Schleswig, Rathausmarkt site, 1070–1140 AD, *RM late* Schleswig, Rathausmarkt site, 1140–1210 AD, *SC* Schleswig, St. Clemens site, 1250–1350 AD

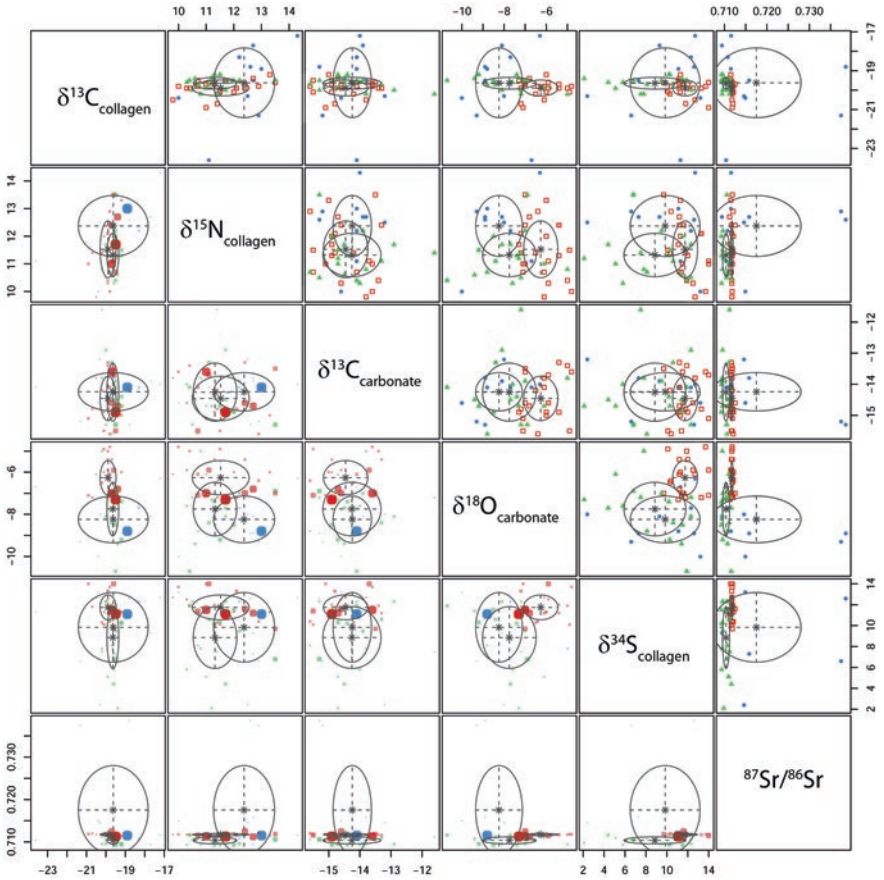


Fig. 9.6 Bivariate scatter plots for the data set of at least adult humans from Haithabu and Schleswig indicating classified GMM clusters (above diagonal; cluster 1, blue dots; cluster 2, red squares; cluster 3, green triangles) and uncertainty of the cluster result (below diagonal; cluster 1, blue; cluster 2, red; cluster 3, green; small symbols, uncertainty <0.1; medium symbols, uncertainty = [0.1, 0.25]; large symbols, uncertainty ≥0.25; color intensity increases with increasing uncertainty). GMM cluster analysis was performed using the "mclust" package. (See Scrucca et al., 2016)

Göhring et al., 2020c) present in almost all individuals of the data set, we can detect a certain freshwater signal in the individuals of cluster 3. This freshwater signal is not only evident in the $\delta^{18}O_{carbonate}$ and $\delta^{34}S_{collagen}$ values but also in the $^{87}Sr/^{86}Sr$ isotope ratios. While the isotopic enrichment in both ^{18}O and ^{34}S could be explained by a marine impact, including the sea spray effect, this would also cause more positive $\delta^{13}C_{carbonate}$ values of the individuals of cluster 3 (see Göhring et al., 2018, 2020c). However, as already mentioned, the individuals do not differ with respect to their $\delta^{13}C_{carbonate}$ values. All individuals of the data set consumed at least some (marine) fish, what comes to no surprise in a coastal environment. The observed discrepancy can be explained by considering the consumption of freshwater fish. $\delta^{18}O$ values in

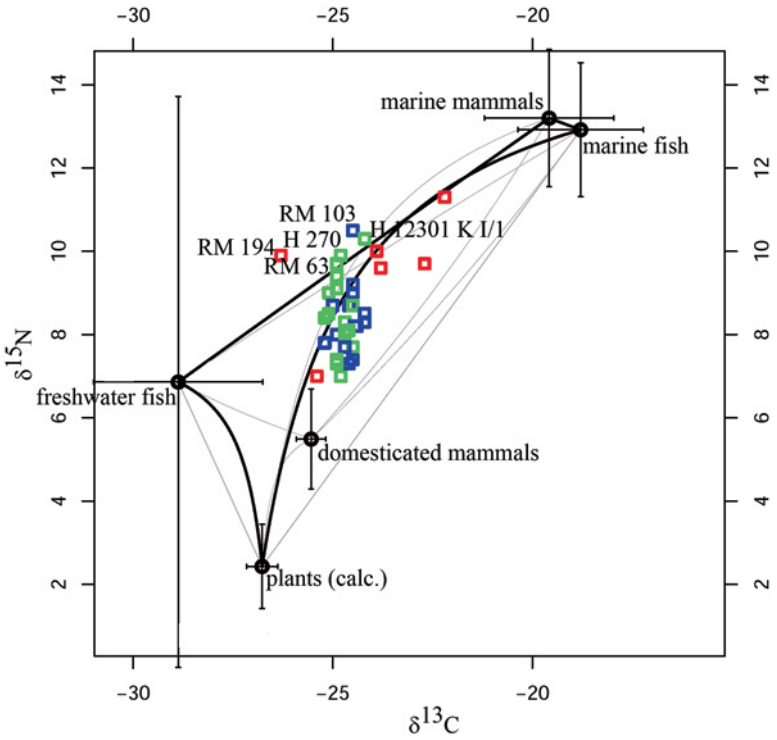


Fig. 9.7 Stable isotope mixing polygon calculated by SISUS indicating the example subset of humans from Haithabu and Schleswig (squares) as well as food sources used for the calculation of the model (black circles; including error bars). The color coding for the humans is according to the GMM clustering results (cluster 1, blue; cluster 2, red; cluster 3, green). Individuals falling outside the polygon are labeled. For information on the food sources as well as the SISUS model for the complete data set for the humans from Haithabu and Schleswig, see Göhring et al. (2020c)

Table 9.7 Mean (μ) and standard deviation (sd) of the data set of at least adult humans from Haithabu and Schleswig

μ	**Cluster 1**	**Cluster 2**	**Cluster 3**
$\delta^{13}C_{collagen}$	−19.63	−19.07	−19.77
$\delta^{15}N_{collagen}$	11.45	12.53	11.54
$\delta^{13}C_{carbonate}$	−14.23	−14.54	−14.51
$\delta^{18}O_{carbonate}$	−7.66	−8.57	−6.07
$\delta^{34}S_{collagen}$	8.24	10.91	11.71
$^{87}Sr/^{86}Sr$	0.71025	0.72074	0.71167
sd	**Cluster 1**	**Cluster 2**	**Cluster 3**
$\delta^{13}C_{collagen}$	0.32	1.44	0.26
$\delta^{15}N_{collagen}$	0.82	1.32	0.94
$\delta^{13}C_{carbonate}$	0.96	0.57	0.76
$\delta^{18}O_{carbonate}$	1.30	1.20	0.81
$\delta^{34}S_{collagen}$	2.95	2.36	1.13
$^{87}Sr/^{86}Sr$	0.00087	0.01253	0.00024

fish reflect the environment in which the fish live, including information on both the latitude and the water temperature (see, e.g., Göhring et al., 2016). Elevated $\delta^{18}O$ values can therefore be due to higher water temperatures in the ponds or lakes from where the freshwater fish might have been caught. $\delta^{34}S_{collagen}$ in individuals living on a terrestrial and/or freshwater diet can reach values of up to 14‰ (Nehlich & Richards, 2009). Strontium incorporated into the skeleton is also largely dependent on diet (Burton & Price, 2013; Lengfelder et al., 2019). In the area under study, the $^{87}Sr/^{86}Sr$ ratio of freshwater can be higher than that of the Schlei inlet as well as the Baltic Sea, further affecting the fish living in the different environments (see also Lahtinen et al., 2021). The consumption of freshwater fish will thus lead to an additional freshwater signal in the bones of the consumers. While higher $^{87}Sr/^{86}Sr$ ratios can also indicate a different origin, the overall identical diet found in the individuals of both clusters 1 and 3 makes a nonlocal origin of the individuals of cluster 3 rather unlikely – in contrast to those individuals of cluster 2 which do not only exhibit different $^{87}Sr/^{86}Sr$ ratios but even a different diet. More positive isotope signals in $\delta^{18}O_{carbonate}$, $\delta^{34}S_{collagen}$, and $^{87}Sr/^{86}Sr$ can thus be explained by a diet containing a certain amount of freshwater fish and at the same time explains why $\delta^{13}C_{carbonate}$ lacks a marine signal. Interestingly, the majority of individuals grouped into cluster 3 (75%) are from Haithabu, with only four individuals from Schleswig (RM 63/145/208, SC 105; see Table 9.6). This indicates dietary shifts from Viking Haithabu to Medieval Schleswig what corresponds to the archaeozoological record (see, e.g., Heinrich, 1987, 1991; Hüster-Plogmann, 2006; Hüster, 1990; Lepiksaar & Heinrich, 1977; Pieper & Reichstein, 1995; Reichstein, 1991; Reichstein & Pieper, 1986).

The results of this example illustrate the spectrum of information provided by GMM cluster analysis, this time combined with isotopic sourcing for palaeodiet reconstruction that was helpful in explaining the GMM results.

9.4 Discussion

Multidimensional stable isotope data sets become more and more common in bioarchaeology. In comparison to one- and two-dimensional data sets, additional variables (depending on the environment and/or the research question) can help to better define and quantify individual diets and detect probably nonlocal individuals. It needs to be emphasized, though, that modeling is not an end in itself and that the choice of the isotopic systems should correspond to the specific research question(s) rather than being based on the mere possibility to measure additional stable isotope ratios.

Multidimensional isotope data contain a lot of different information. However, it is possible that certain isotopic systems disturb the similarity search, as demonstrated in Example 9.1. This has to be controlled by using, for example, ARI. Also, in the case of archaeological material, it might be necessary to reduce the number of analyzed isotopic systems due to a lack in sample material. In such a case, feature ranking can help to detect the isotopic system(s) which could be omitted from

analysis without losing (too) much information (see Göhring et al., 2020b; Mauder et al., 2019).

Using a multidimensional approach is usually preferable. However, for the interpretation of multi-isotope fingerprints, a data mining approach such as GMM cluster analysis is required. While it is theoretically possible to create three- or even four-dimensional plots, the readability of such illustrations is quite poor. As demonstrated in this paper, cluster analysis still allows the use of simple bivariate plots including the information of all dimensions under study by illustration of the allocated clusters using different colors and/or symbols.

Contextually, it is of major importance to always keep in mind that any cluster analysis requires a comprehensive subsequent interpretation. A cluster captures all multidimensional data sets that are similar to each other and dissimilar from the others, but this result needs to be deciphered in terms of the classified individuals, cluster mean or standard deviation, clustering uncertainty, and the general information provided by a stable isotope ratio. Choice of stable isotope systems included in the clustering procedure is crucial and thus probably the most important step within the KDD approach. As illustrated for the transalpine example, clustering results markedly change and become much more meaningful after removal of $\delta^{18}O_{phosphate}$. Mainly representative of the water source (e.g., Kirsanow & Tuross, 2011), $\delta^{18}O_{precipitation}$ and accordingly $\delta^{18}O_{phosphate}$, are too variable across the whole transect and thus do not allow distinction between the northern, inneralpine, and southern regions. In other regions, for example, at coastal sites such as Haithabu and Schleswig, stable oxygen isotopes can include indispensable information. At coastal sites, $\delta^{18}O$ can be of particular importance for clustering data of both terrestrial vertebrates, for example, affected by the sea spray effect (see, e.g., Göhring et al., 2018, 2019, 2020c) and aquatic vertebrates, where $\delta^{18}O$ reflects parameters of the ambient water body. The amount of information hidden in the different isotopic systems as well as in the combination of isotopic systems can be controlled by the performance of a feature ranking method. It can therefore be advisable to create several GMM cluster models with more or fewer isotopic systems, depending on the research question(s). In addition, preprocessing or transformation of the data might be helpful to identify certain underlying structures.

Moreover, the calculation of clustering uncertainty (or probability) is a big advantage of the probabilistic GMM approach compared with other methods such as k-means which group data points to centroids, starting with an initial set of centroids according to their (Euclidean) distance from the centroids, but without calculating a clustering probability. The centroids are updated, and the procedure is repeated stepwise until the convergence criterion is fulfilled, that is, until the centroids stay the same (for an overview, see, e.g., Reddy & Vinzamuri, 2014). Due to the dependence from the initial set of centroids, k-means does not necessarily result in the best solution but can converge to another solution. In contrast, the EM algorithm of the GMM approach does always result in a BIC optimal cluster model. While different other clustering approaches could actually be applied to stable

isotope data, GMM resulted in the most convincing results for the data sets presented here.

Clustering uncertainty is of special contextual importance since individuals with a low cluster assignment ($p <$ ca. 0.7) only partly fit into the overall data structure. Primarily nonlocal individuals, individuals with mixed isotope signals due to intra-life migration, or individuals consuming a somehow different diet do not necessarily fall into a mathematical outlier group, although possible, but become visible by a low clustering probability. Accordingly, those individuals need special attention with regard to the research question. This kind of information is usually invisible in traditional data analysis methods (e.g., bivariate plots, PCA).

9.5 Conclusion

We presented GMM cluster analysis as a new and promising tool for the evaluation of the structure of multidimensional isotope data sets and its contextual interpretation. This method can result in additional (and also entirely new) knowledge, very much in the spirit of KDD. In accordance with the KDD process, several approaches such as starting with a subset of the data (e.g., during a feature ranking approach) or additional information of other methods of analysis (e.g., isotopic sourcing) may be required to maximize the information hidden in the data. As demonstrated by three examples related to different research questions, GMM gives clues to, for example, provenience, diet, or palaeoecological studies in diverse and even highly complex environments based on multi-isotope fingerprints, where traditional data analysis methods, such as bivariate plots or PCA, will remain – in many cases – insufficient.

An important perspective with regard to multidimensional stable isotope data sets is the capability of GMM to potentially identify individuals with mixed isotopic signals due to, for example, an intra-life migration event. This is a specific problem when bone needs to be analyzed instead of tooth enamel, usually in human cremations where enamel is hardly preserved. Individuals with mixed isotopic ratios will exhibit fingerprints that fall in between two clusters and are conspicuous because of rather high clustering uncertainties. With regard to provenance studies, such individuals should be considered as potential nonlocals.

As is the case for all modeling and computer-aided procedures, it cannot be emphasized enough that while also GMM quickly and always results in a clustering outcome, the results necessitate a careful and in-depth interpretation with regard to the research question(s) and the archaeological context. Moreover, we would like to draw the reader's attention to the obligatory data cleaning procedure. For example, data that do not meet the quality criteria commonly agreed upon (such as the measurement error) have to be excluded prior to the clustering approach to obtain robust and valid results.

References

Aggarwal, C. C. (2014). An introduction to cluster analysis. In C. C. Aggarwal & C. K. Reddy (Eds.), *Data clustering: Algorithms and applications* (pp. 1–27). Chapman, Hall/CRC. https://doi.org/10.1201/9781315373515-1

Aggarwal, C. C. (2015). *Data mining: The textbook*. Springer. https://doi.org/10.1007/978-3-319-14142-8

Alonzi, E., Pacheco-Forés, S. I., Gordon, G. W., Kuijt, I., & Knudson, K. J. (2020). New understandings of the sea spray effect and its impact on bioavailable radiogenic strontium isotope ratios in coastal environments. *Journal of Archaeological Science: Reports, 33*, 102462. https://doi.org/10.1016/j.jasrep.2020.102462

Artioli, G., Angelini, I., Nimis, P., & Villa, I. M. (2016). A lead-isotope database of copper ores from the southeastern Alps: A tool for the investigation of prehistoric copper metallurgy. *Journal of Archaeological Science, 75*, 27–39. https://doi.org/10.1016/j.jas.2016.09.005

Bensmail, H., Celeux, G., Raftery, A. E., & Robert, C. P. (1997). Inference in model-based cluster analysis. *Statistics and Computing, 7*, 1–10. https://doi.org/10.1023/a:1018510926151

Bowen, G. J., West, J. B., Vaughn, B. H., Dawson, T. E., Ehleringer, J. R., Fogel, M. L., Hobson, K., Hoogewerff, J., Kendall, C., Lai, C.-T., Miller, C. C., Noone, D., Schwarcz, H., & Still, C. J. (2009). Isoscapes to address large-scale earth science challenges. *Eos, 90*, 109–110. https://doi.org/10.1029/2009eo130001

Burton, J. H., & Price, T. D. (2013). Seeking the local $^{87}Sr/^{86}Sr$ ratio to determine geographic origins of humans. In *Archaeological chemistry VIII* (ACS symposium series 1147) (pp. 308–320). American Chemical Society. https://doi.org/10.1021/bk-2013-1147.ch018

Deng, H., & Han, J. (2014). Probabilistic models for clustering. In C. C. Aggarwal & C. K. Reddy (Eds.), *Data clustering: Algorithms and applications* (pp. 61–86). Chapman, Hall/CRC. https://doi.org/10.1201/9781315373515-3

DeNiro, M. J. (1985). Postmortem preservation and alteration of in vivo bone collagen isotope ratios in relation to palaeodietary reconstruction. *Nature, 317*(6040), 806–809.

Doppler, S., Vohberger, M., Von Carnap-Bornheim, C., Peters, J., & Grupe, G. (2010). Biodiversity of archaeological fauna in the estuarine palaeoecosystem of the Schlei fjord, Northern Germany: Isotopic evidence. *Documenta Archaeobiologiae, 8*, 21–70.

Drivelos, S. A., & Georgiou, C. A. (2012). Multi-element and multi-isotope-ratio analysis to determine the geographical origin of foods in the European Union. *Trends in Analytical Chemistry, 40*, 38–51. https://doi.org/10.1016/j.trac.2012.08.003

Erhardt, E. B. (2014). *SISUS: Stable isotope sourcing using sampling* [Online]. Available https://cran.r-project.org/web/packages/sisus/

Fayyad, U. M. (1996). Data mining and knowledge discovery: Making sense out of data. *IEEE Expert: Intelligent Systems and Their Applications, 11*, 20–25. https://doi.org/10.1109/64.539013

Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996a). From data mining to knowledge discovery in databases. *AI Magazine, 17*, 37–54.

Fayyad, U. M., Piatetsky-Shapiro, G., & Smyth, P. (1996b). From data mining to knowledge discovery: An overview. In U. M. Fayyad, G. Piatetsky-Shapiro, P. Smyth, & R. Uthurusamy (Eds.), *Advances in knowledge discovery and data mining* (pp. 37–54). AAAI Press.

Filzmoser, P., & Gschwandtner, M. (2018). *mvoutlier: Multivariate outlier detection based on robust methods* [Online]. Available http://CRAN.R-project.org/package=mvoutlier

Fraley, C., & Raftery, A. E. (1998). How many clusters? Which clustering method? Answers via model-based cluster analysis. *The Computer Journal, 41*, 578–588. https://doi.org/10.1093/comjnl/41.8.578

Fraley, C., & Raftery, A. E. (2002). Model-based clustering, discriminant analysis, and density estimation. *Journal of the American Statistical Association, 97*, 611–631. https://doi.org/10.2307/3085676

Frawley, W. J., Piatetsky-Shapiro, G., & Matheus, C. J. (1992). Knowledge discovery in databases: An overview. *AI Magazine, 13*, 57–70.

Fukuda, K. (2005). *cddlib Reference Manual*. Available online ftp://ftp.math.ethz.ch/users/fukuda/cdd/cddlibman/cddlibman.html

Fukuda, K., & Prodon, A. (1996). Double description method revisited. In M. Deza, R. Euler, & I. Manoussakis (Eds.), *8th Franco-Japanese and 4th Franco-Chinese Conference Brest, France*, July 3–5, 1995 (pp. 91–111). Springer. https://doi.org/10.1007/3-540-61576-8_77

Göhring, A. (2019). *Anwendung KDD-basierter Methoden zur Interpretation multi-dimensionaler Isotopen-Fingerabdrücke.* Dissertation, Ludwig-Maximilians-Universität München. https://doi.org/10.5282/edoc.24902

Göhring, A. (2020a). *R script for Gaussian Mixture Model (GMM) cluster analysis*. Available online https://www.for1670-transalpine.uni-muenchen.de/tools

Göhring, A. (2020b). Morphological and histological survey of human remains from Tepe 5. In K. Kaniuth (Ed.), *Karacamirli – Tepe 5. A multi-period necropolis in Western Azerbaijan* (Münchener Abhandlungen zum Alten Orient 6) (pp. 163–216). PeWe-Verlag

Göhring, A., Mauder, M., Kröger, P., & Grupe, G. (2016). Using Gaussian Mixture Model clustering for multi-isotope analysis of archaeological fish bones for palaeobiodiversity studies. *Rapid Communications in Mass Spectrometry, 30*, 1349–1360. https://doi.org/10.1002/rcm.7573

Göhring, A., Mauder, M., Vohberger, M., Nehlich, O., von Carnap-Bornheim, C., Hilberg, V., Kröger, P., & Grupe, G. (2018). Palaeobiodiversity research based on stable isotopes: Correction of the sea spray effect on bone carbonate $\delta^{13}C$ and $\delta^{18}O$ by Gaussian Mixture Model clustering. *Palaeogeography, Palaeoclimatology, Palaeoecology, 290*, 673–686. https://doi.org/10.1016/j.palaeo.2017.11.057

Göhring, A., Mauder, M., Kröger, P., Mayr, C., von Carnap-Bornheim, C., Hilberg, V., & Grupe, G. (2019). Evidence for sea spray effect on oxygen stable isotopes in bone phosphate – Approximation and correction using Gaussian Mixture Model clustering. *Science of the Total Environment, 673*, 668–684. https://doi.org/10.1016/j.scitotenv.2019.04.072

Göhring, A., Hölzl, S., Joachimski, M. M., Mayr, C., & Grupe, G. (2020a). Multi-isotope fingerprint of humans, animals, and environmental samples from Karacamirli. In K. Kaniuth (Ed.), *Karacamirli – Tepe 5. A multi-period necropolis in Western Azerbaijan* (Münchener Abhandlungen zum Alten Orient 6) (pp. 217–245). PeWe-Verlag.

Göhring, A., Mauder, M., Kröger, P., & Grupe, G. (2020b). Which isotopes should we choose? Entropy-based feature selection enables ranking of stable isotopes in archaeofaunal material. *Rapid Communications in Mass Spectrometry, 34*, e8692. https://doi.org/10.1002/rcm.8692

Göhring, A., Mayr, C., & Grupe, G. (2020c). Sea spray correction in $\delta^{13}C_{carbonate}$, $\delta^{18}O_{carbonate}$, $\delta^{18}O_{phosphate}$, and $\delta^{34}S_{collagen}$ values of coastal humans – A methodological approach. *Science of the Total Environment, 744*, 140907. https://doi.org/10.1016/j.scitotenv.2020.140907

Göhring, G., Hölzl, S., Mayr, C., & Strauss, H. (2023). Identification and quantification of the sea spray effect on isotopic systems in α-cellulose ($\delta^{13}C$, $\delta^{18}O$), total sulfur ($\delta^{34}S$), and $^{87}Sr/^{86}Sr$ of European beach grass (*Ammophila arenaria*, L.) in a greenhouse experiment. *Science of the Total Environment, 856*(1), 158840. https://doi.org/10.1016/j.scitotenv.2022.158840

Grupe, G. (2014). Application of isotopic mixing models for palaeodietary and paleoecological studies. *Anthropologischer Anzeiger, 71*, 21–39. https://doi.org/10.1127/0003-5548/2014/0375

Grupe, G., von Carnap-Bornheim, C., & Becker, C. (2013). Rise and fall of a medieval trade centre: Economic change from Viking Haithabu to Medieval Schleswig revealed by stable isotope analysis. *European Journal of Archaeology, 16*, 137–166. https://doi.org/10.1179/1461957112Y.0000000021

Grupe, G., Hölzl, S., Mayr, C., & Söllner, F. (2017). The concept of isotopic landscapes: Modern ecogeochemistry versus bioarchaeology. In G. Grupe, A. Grigat, & G. C. McGlynn (Eds.), *Across the Alps in prehistory: Isotopic mapping of the Brenner passage by bioarchaeology* (pp. 27–48). Springer. https://doi.org/10.1007/978-3-319-41550-5_2

Grupe, G., Klaut, D., Mauder, M., Kröger, P., Lang, A., Mayr, C., & Söllner, F. (2018). Multi-isotope provenancing of archaeological skeletons including cremations in a reference area of the European Alps. *Rapid Communications in Mass Spectrometry, 32*, 1711–1727. https://doi.org/10.1002/rcm.8218

Grupe, G., Klaut, D., Otto, L., Mauder, M., Lohrer, J., Kröger, P., & Lang, A. (2020). The genesis and spread of the early Fritzens-Sanzeno culture (5th/4th cent. BCE) – Stable isotope analysis of cremated and uncremated skeletal finds. *Journal of Archaeological Science: Reports, 29*, 102121. https://doi.org/10.1016/j.jasrep.2019.102121

Haack, U. K., Heinrichs, H., Gutsche, F. H., & Plessow, K. (2003). The isotopic composition of anthropogenic Pb in soil profiles of Northern Germany: Evidence for pollutant Pb from a continent-wide mixing system. *Water, Air, and Soil Pollution, 150*, 113–134. https://doi.org/10.1023/A:1026142501593

Hagl, P., Becker, C., & Grupe, G. (2013). Establishment of dairying economy in the Bavarian Neolithic revealed by isotopic foodwebs. *Documenta Archaeobiologiae, 11*, 79–93.

Harbeck, M., & Grupe, G. (2009). Experimental chemical degradation compared to natural diagenetic alteration of collagen: Implications for collagen quality indicators for stable isotope analysis. *Archaeological and Anthropological Sciences, 1*, 43–57. https://doi.org/10.1007/s12520-009-0004-5

Heinrich, D. (1987). Untersuchungen an mittelalterlichen Fischresten aus Schleswig. Ausgrabung Schild 1971-1975. In V. Vogel (Ed.), *Ausgrabungen in Schleswig: Berichte und Studien 6*. Karl Wachholtz.

Heinrich, D. (1991). *Untersuchungen an Skelettresten wildlebender Säugetiere aus dem mittelalterlichen Schleswig*. Karl Wachholtz Verlag.

Horikoshi, M., & Tang, Y. (2016). *ggfortify: Data visualization tools for statistical analysis results*. Available online https://CRAN.R-project.org/package=ggfortify

Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification, 2*, 193–218. https://doi.org/10.1007/bf01908075

Hüster, H. (1990). Untersuchungen an Skelettresten von Rindern, Schafen, Ziegen und Schweinen aus dem mittelalterlichen Schleswig. Ausgrabung Schild 1971-1975. In V. Vogel (Ed.), *Ausgrabungen in Schleswig: Berichte und Studien 8*. Karl Wachholtz.

Hüster-Plogmann, H. (2006). Untersuchungen an den Skelettresten von Säugetieren und Vögeln aus dem Hafen von Haithabu. In K. Schietzel (Ed.), *Berichte über die Ausgrabungen in Haithabu: Bericht 35* (pp. 25–156). Wachholtz Verlag.

Jay, M., Montgomery, J., Nehlich, O., Towers, J. R., & Evans, J. (2013). British Iron Age chariot burials of the Arras culture: A multi-isotope approach to investigating mobility levels and subsistence practices. *World Archaeology, 45*, 473–491. https://doi.org/10.1080/00438243.2013.820647

King, C. L., Buckley, H. R., Petchey, P., Roberts, P., Zech, J., Kinaston, R., Collins, C., Kardailsky, O., Matisoo-Smith, E., & Nowell, G. (2021). An isotopic and genetic study of multicultural colonial New Zealand. *Journal of Archaeological Science, 128*, 105337. https://doi.org/10.1016/j.jas.2021.105337

Kirsanow, K., & Tuross, N. (2011). Oxygen and hydrogen isotopes in rodent tissues: Impact of diet, water and ontogeny. *Palaeogeography, Palaeoclimatology, Palaeoecology, 310*, 9–16. https://doi.org/10.1016/j.palaeo.2011.03.022

Krueger, H. W., & Sullivan, C. H. (1984). Models for carbon isotope fractionation between diet and bone. In J. R. Turnlund & P. E. Johnson (Eds.), *Stable isotopes in nutrition* (ACS symposium series 258) (pp. 205–222). American Chemical Society.

Kusaka, S., Ando, A., Nakano, T., Yumoto, T., Ishimaru, E., Yoneda, M., Hyodo, F., & Katayama, K. (2009). A strontium isotope analysis on the relationship between ritual tooth ablation and migration among the Jomon people in Japan. *Journal of Archaeological Science, 36*, 2289–2297. https://doi.org/10.1016/j.jas.2009.06.013

Lafoon, J. E., Hoogland, M. L. P., Davies, G. R., & Hofmann, C. L. (2019). A multi-isotope investigation of human and dog mobility and diet in die Pro-Colonial Antilles. *Environmental Archaeology, 24*, 132–148. https://doi.org/10.1080/14614103.2017.1322831

Lahtinen, M., Arppe, L., & Nowell, G. (2021). Source of strontium in archaeological mobility studies – Marine diet contribution to the isotopic composition. *Archaeological and Anthropological Sciences, 13*, 1. https://doi.org/10.1007/s12520-020-01240-w

Lengfelder, F., Grupe, G., Stallauer, A., Huth, R., & Söllner, F. (2019). Modelling strontium isotopes in past biospheres – Assessment of bioavailable $^{87}Sr/^{86}Sr$ ratios in local archaeological vertebrates based on environmental signatures. *Science of the Total Environment, 648*, 236–252. https://doi.org/10.1016/j.scitotenv.2018.08.014

Lepiksaar, J., & Heinrich, D. (1977). Untersuchungen an Fischresten aus der frühmittelalterlichen Siedlung Haithabu. In K. Schietzel (Ed.), *Berichte über die Ausgrabungen in Haithabu 10* (pp. 9–122). Karl Wachholtz.

Madgwick, R., Lamb, A. L., Sloane, H., Nederbragt, A. J., Parker Pearson, M., & Evans, J. (2019). Multi-isotope analysis reveals that feasts in the Stonehenge environs and across Wessex drew people and animals from throughout Britain. *Science Advances, 5*(3), eaau6078. https://doi.org/10.1126/sciadv.aau6078

Mauder, M. (2017). *Analyzing complex data using domain constraints*. Dissertation, Ludwig-Maximilians-Universität. München. https://doi.org/10.5282/edoc.20890

Mauder, M., Ntoutsi, E., Kröger, P., & Kriegel, H.-P. (2016a). Towards predicting places of origin from isotopic fingerprints: A case study on the mobility of people in the Central European Alps. In G. Grupe & G. C. McGlynn (Eds.), *Isotopic landscapes in bioarchaeology* (pp. 221–233). Springer. https://doi.org/10.1007/978-3-662-48339-8_13

Mauder, M., Ntoutsi, E., Kröger, P., Mayr, C., Grupe, G., Toncala, A., & Hölzl, S. (2016b). Applying data mining methods for the analysis of stable isotope data in bioarchaeology. In *12th IEEE international conference on e-science, e-science 2016* (pp. 233–242). IEEE. https://doi.org/10.1109/eScience.2016.7870904

Mauder, M., Bobkova, Y., & Ntoutsi, E. (2016c). GMMbuilder – User-driven discovery of clustering structure for bioarchaeology. In B. Berendt, B. Bringmann, É. Fromont, G. Garriga, P. Miettinen, N. Tatti, & V. Tresp (Eds.), *Machine learning and knowledge discovery in databases: European conference, ECML PKDD 2016, Riva del Garda, Italy, September 19–23, 2016, Proceedings, Part III* (pp. 8–11). Springer. https://doi.org/10.1007/978-3-319-46131-1_2

Mauder, M., Ntoutsi, E., Kröger, P., & Kriegel, H.-P. (2017). The isotopic fingerprint: New methods of data mining and similarity search. In G. Grupe, A. Grigat, & G. C. McGlynn (Eds.), *Across the Alps in prehistory: Isotopic mapping of the Brenner passage by bioarchaeology* (pp. 105–126). Springer. https://doi.org/10.1007/978-3-319-41550-5_5

Mauder, M., Ntoutsi, E., Kröger, P., Mayr, C., Toncala, A., Hölzl, S., & Grupe, G. (2019). Significance and limitations of stable oxygen isotope ratios in the apatite phosphate of archaeological vertebrate finds for provenance analysis in an Alpine reference region. *Archaeometry, 61*, 194–210. https://doi.org/10.1111/arcm.12399

Metzner-Nebelsick, C., Lang, A., Sommer, C. S., & Steidl, B. (2017). Transalpine mobility and trade since the Mesolithic. In G. Grupe, A. Grigat, & G. C. McGlynn (Eds.), *Across the Alps in prehistory: Isotopic mapping of the Brenner passage by bioarchaeology* (pp. 1–26). Springer. https://doi.org/10.1007/978-3-319-41550-5_1

Nehlich, O., & Richards, M. (2009). Establishing collagen quality criteria for sulphur isotope analysis of archaeological bone collagen. *Archaeological and Anthropological Sciences, 1*, 59–75. https://doi.org/10.1007/s12520-009-0003-6

Phillips, D. L., Inger, R., Bearhop, S., Jackson, A. L., Moore, J. W., Parnell, A. C., Semmens, B. X., & Ward, E. J. (2014). Best practices for use of stable isotope mixing models in food-web studies. *Canadian Journal of Zoology, 92*, 823–835. https://doi.org/10.1139/cjz-2014-0127

Piatetsky-Shapiro, G. (1990). Knowledge discovery in real databases: A report on the IJCAI-89 workshop. *AI Magazine, 11*, 68–70.

Pieper, H., & Reichstein, H. (1995). Untersuchungen an Skelettresten von Vögeln aus dem mittelalterlichen Schleswig. In V. Vogel (Ed.), *Ausgrabungen in Schleswig: Berichte und Studien 11* (pp. 9–113). Karl Wachholtz Verlag.

Price, T. D., Frei, R., Bäckström, Y., Frei, K. M., & Ingvarsson-Sundstrom, A. (2017). Origins of inhabitants from the 16th century Sala (Sweden) silver mine cemetery – A lead isotope perspective. *Journal of Archaeological Science, 80*, 1–13. https://doi.org/10.1016/j.jas.2017.01.013

Price, T. D., Arcini, C., Gustin, I., Drenzel, L., & Kalmring, S. (2018). Isotopes and human burials at Viking Age Birka and the Mälaren region, east Central Sweden. *Journal of Anthropological Archaeology, 49*, 19–38. https://doi.org/10.1016/j.jaa.2017.10.002

Price, T. D., Moiseyev, V., & Grigoreva, N. (2019). Vikings in Russia: Origins of the medieval inhabitants of Staraya Ladoga. *Archaeological and Anthropological Sciences, 11*, 6093–6109. https://doi.org/10.1007/s12520-019-00897-2

R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.

Reddy, C. K., & Vinzamuri, B. (2014). A survey of partitional and hierarchical clustering algorithms. In C. C. Aggarwal & C. K. Reddy (Eds.), *Data clustering: Algorithms and applications* (pp. 87–110). Chapman, Hall/CRC. https://doi.org/10.1201/9781315373515-4

Reichstein, H. (1974). Ergebnisse und Probleme von Untersuchungen an Wildtieren aus Haithabu (Ausgrabung 1963-1964). In K. Schietzel (Ed.), *Berichte über die Ausgrabungen in Haithabu: Bericht 7* (pp. 103–144). Karl Wachholtz.

Reichstein, H. (1984). Die Tierwelt im Bereich von Haithabu. In H. Jankuhn, K. Schietzel, & H. Reichstein (Eds.), *Archäologische und naturwissenschaftliche Untersuchungen an ländlichen und frühstädtischen Siedlungen im deutschen Küstengebiet vom 5. Jahrhundert v. Chr. bis zum 11. Jahrhundert n. Chr. Band 2: Handelsplätze des frühen und hohen Mittelalters* (pp. 78–82). Acta humaniora.

Reichstein, H. (1991). Die wildlebenden Säugetiere von Haithabu (Ausgrabungen 1966-1969 und 1979-1980). In K. Schietzel (Ed.), *Berichte über die Ausgrabungen in Haithabu: Bericht 30*. Karl Wachholtz.

Reichstein, H., & Pieper, H. (1986). Untersuchungen an Skelettresten von Vögeln aus Haithabu (Ausgrabung 1966-1969). In K. Schietzel (Ed.), *Berichte über die Ausgrabungen in Haithabu: Bericht 22*. Karl Wachholtz.

Requate, H. (1956). Die Jagdtiere in den Nahrungsresten einiger frühgeschichtlicher Siedlungen in Schleswig-Holstein: Ein Beitrag zur Faunengeschichte des Landes. *Schriften des Naturwissenschaftlichen Vereins für Schleswig-Holstein, 28*, 21–41.

RStudio Team. (2020). *RStudio: Integrated development for R.* Available online http://www.rstudio.com/

Scorrer, J., Faillace, K. E., Hildred, A., Nedergragt, A. L., Andersen, M. B., Millet, M.-A., Lamb, A. L., & Madgwick, R. (2021). Diversity aboard a Tudor warship: Investigating the origins of the *Mary Rose* crew using multi-isotope analysis. *Royal Society Open Science, 8*, 202106. https://doi.org/10.1098/rsos.202106

Scrucca, L., Fop, M., Murphy, T. B., & Raftery, A. E. (2016). mclust 5: Clustering, classification and density estimation using Gaussian finite mixture models. *The R Journal, 8*, 289–317. https://doi.org/10.32614/RJ-2016-021

Shaw, H., Montgomery, J., Redfern, G., Gowland, R., & Evans, J. (2016). Identifying migrants in Roman London using lead and strontium stable isotopes. *Journal of Archaeological Science, 66*, 57–68. https://doi.org/10.1016/j.jas.2015.12.001

Smith, R. L. (1984). Efficient Monte Carlo procedures for generating points uniformly distributed over bounded regions. *Operations Research, 32*, 1296–1308.

Snoeck, C. (2014). Impact of strontium sea spray effect on the isotopic ratio (87Sr/86Sr) of plants in coastal Ireland. *Quaternary Newsletter, 134*, 37–39.

Toncala, A., Söllner, F., Mayr, C., Hölzl, S., Heck, K., Wycisk, D., & Grupe, G. (2017). Isotopic map of the Inn-Eisack-Adige-Brenner passage and its application to prehistoric human cremations. In G. Grupe, A. Grigat, & G. C. McGlynn (Eds.), *Across the Alps in prehistory: Isotopic mapping of the Brenner passage by bioarchaeology* (pp. 127–228). Springer.

Tschetsch, L., Mussauer, A., Mauder, M., Lohrer, J., Kröger, P., Mayr, C., Söllner, F., & Grupe, G. (2020). Multi-isotope fingerprints (O, Sr and Pb) in archaeological animal bone bioapatite: Similarity search and the suitability for provenance analysis in a geologically complex Alpine region. *Archaeometry, 62*, 35–52. https://doi.org/10.1111/arcm.12550

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer. https://doi.org/10.1007/978-0-387-98141-3

Xiong, H., & Li, Z. (2014). Clustering validation measures. In C. C. Aggarwal & C. K. Reddy (Eds.), *Data clustering: Algorithms and applications* (pp. 571–605). Chapman, Hall/CRC. https://doi.org/10.1201/9781315373515-23